Stratospheric Ozone and the Ozone Hole
EES202
Handouts

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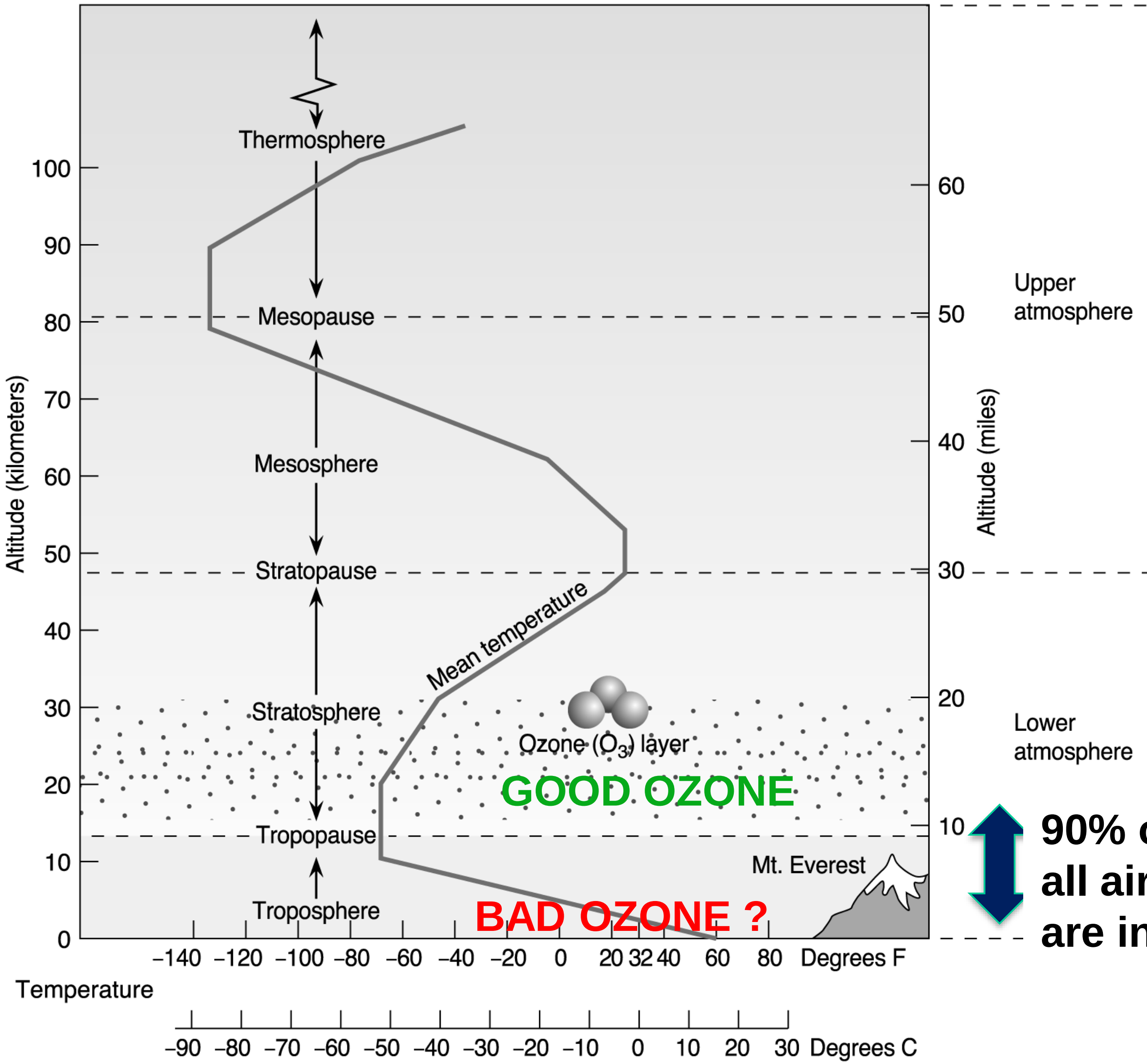
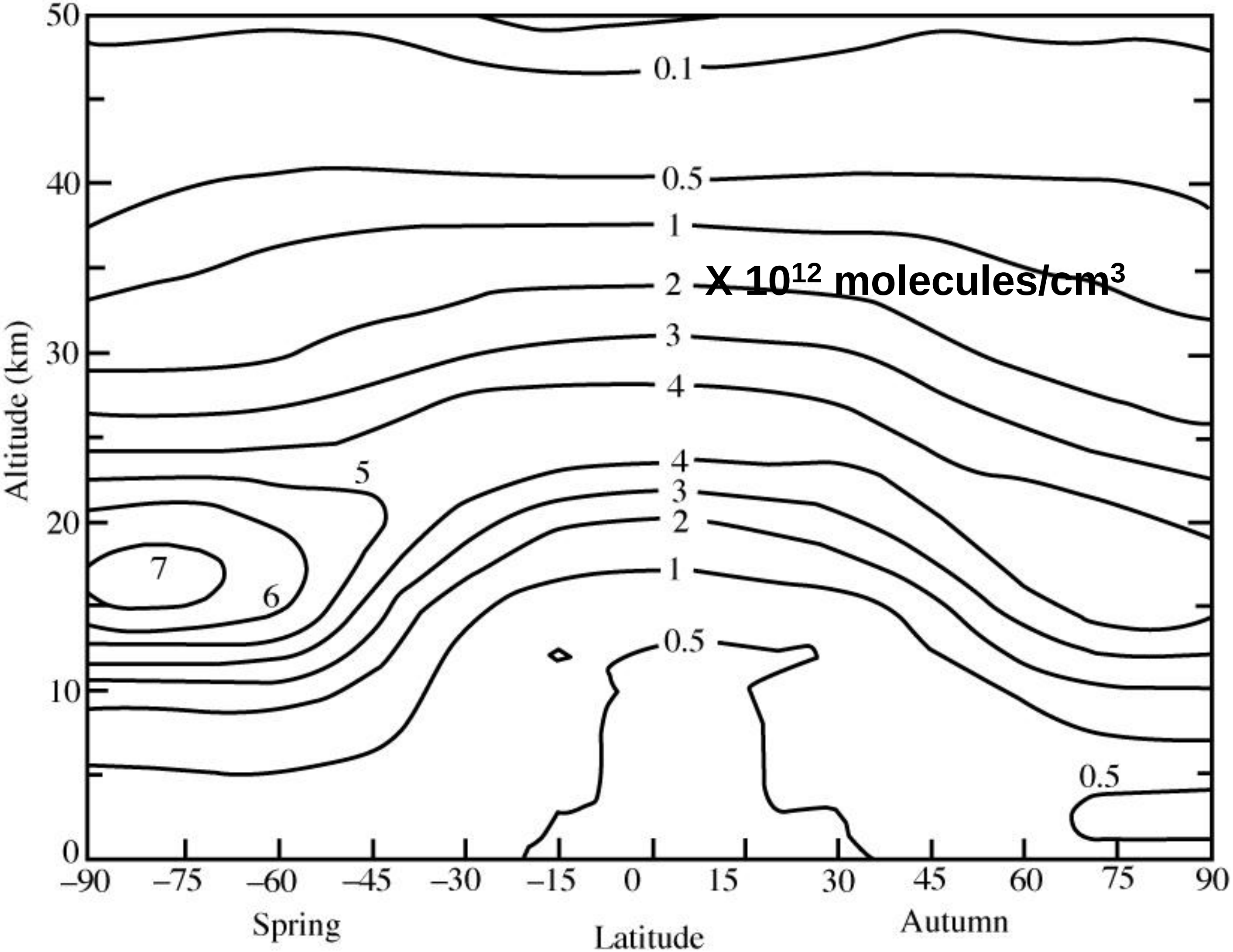
The atmosphere as a reactor (T, P changes)



90% of all air molecules are in the troposph

THE NATURAL OZONE LAYER

Based on ozonesonde observations in the 1970s



Stratospheric Chemistry: Driven by Chemistry of Ozone

• Ozone was discovered in 1839 by Schoenbein but its importance as an atmospheric constituent was established only after 1860

• French scientists Fabry & Buisson used UV measurements to estimate that if brought down to the surface of the Earth, the Ozone layer would produce a layer 3 mm thick (at STP)

• In 1930 British scientist Chapman proposed that ozone is continually produced in the atmosphere by a cycle initiated by photolysis of oxygen in the upper stratosphere

•However Chapman mechanism overestimated Ozone & only in 1970 after Crutzen elucidated the role of NO_x did more accurate understanding of Ozone chemistry evolve

CHAPMAN MECHANISM FOR STRATOSPHERIC OZONE

O_2 Bond energy 498 kJ/mol

$$(R1) \qquad O_2 + h\nu \rightarrow O + O \qquad (\lambda < 240 \text{ nm})$$

O in ground triplet, O (^3P)

$$(R2) \qquad O + O_2 + M \rightarrow O_3 + M$$

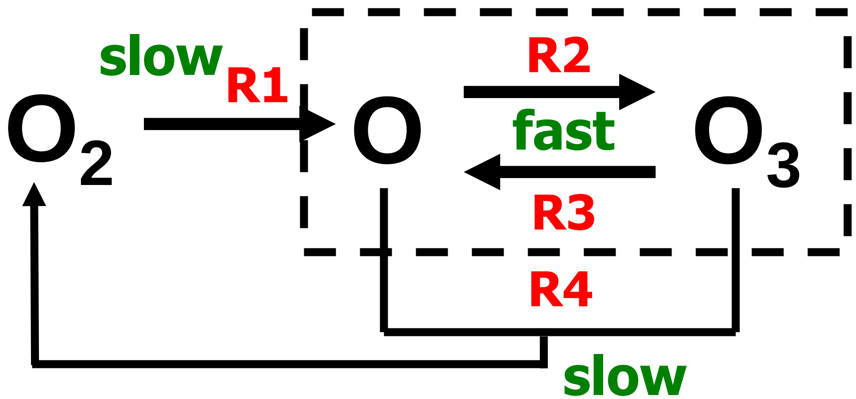
O_3 has weaker bonds than O_2

$$(R3) \qquad O_3 + h\nu \rightarrow O_2 + O \qquad (\lambda < 320 \text{ nm})$$

O in excited singlet, O (^1D)

$$(R4) \qquad O_3 + O \rightarrow 2O_2$$

RATE CONSTANTS FOR R1-R4 HAVE BEEN MEASURED IN LABORATORY & FOUND TO BE:



Odd oxygen *family*
$[O_x] = [O_3] + [O]$
WHICH IS CONSUMED BY R4 AND PRODUCED BY R1

SCHEMATIC OF CHAPMAN MECHANISM

STEADY-STATE ANALYSIS OF CHAPMAN MECHANISM

What is the Lifetime of O atoms in Chapman mechanism ?

$$\tau_{\mathrm{O}} = \frac{[\mathrm{O}]}{k_2[\mathrm{O}][\mathrm{O}_2][\mathrm{M}]+k_4[\mathrm{O}_3][\mathrm{O}]} \approx \frac{1}{k_2 C_{\mathrm{O}2} n_a^2} < 1\ \mathrm{s}$$

Short enough to assume chemical steady state for O (loss = prod) as Pressure, T, Radiation & ozone abundance won't change much in t ~ 1s :

$$R2 = R3 \Rightarrow k_2[\mathrm{O}][\mathrm{O}_2][\mathrm{M}]=k_3[\mathrm{O}_3] \Rightarrow \frac{[\mathrm{O}]}{[\mathrm{O}_3]} = \frac{k_3}{k_2 C_{O2} n_a^2} \ll 1$$

$\Rightarrow [\mathrm{O}_x] \approx [\mathrm{O}_3]$ **…so O_3 makes up most of the O_x budget**

Lifetime of O_x: From R4; 2 coz 2 O_X used

$$\tau_{\mathrm{Ox}} = \frac{[\mathrm{O}_x]}{2k_4[\mathrm{O}_3][\mathrm{O}]} \approx \frac{1}{2k_4[\mathrm{O}]}$$

Assuming Steady state for O_x :

$$\frac{d}{dt}[O_x] = \frac{d}{dt}([O_3]+[O]) = \frac{d}{dt}[O_3]+\frac{d}{dt}[O]$$

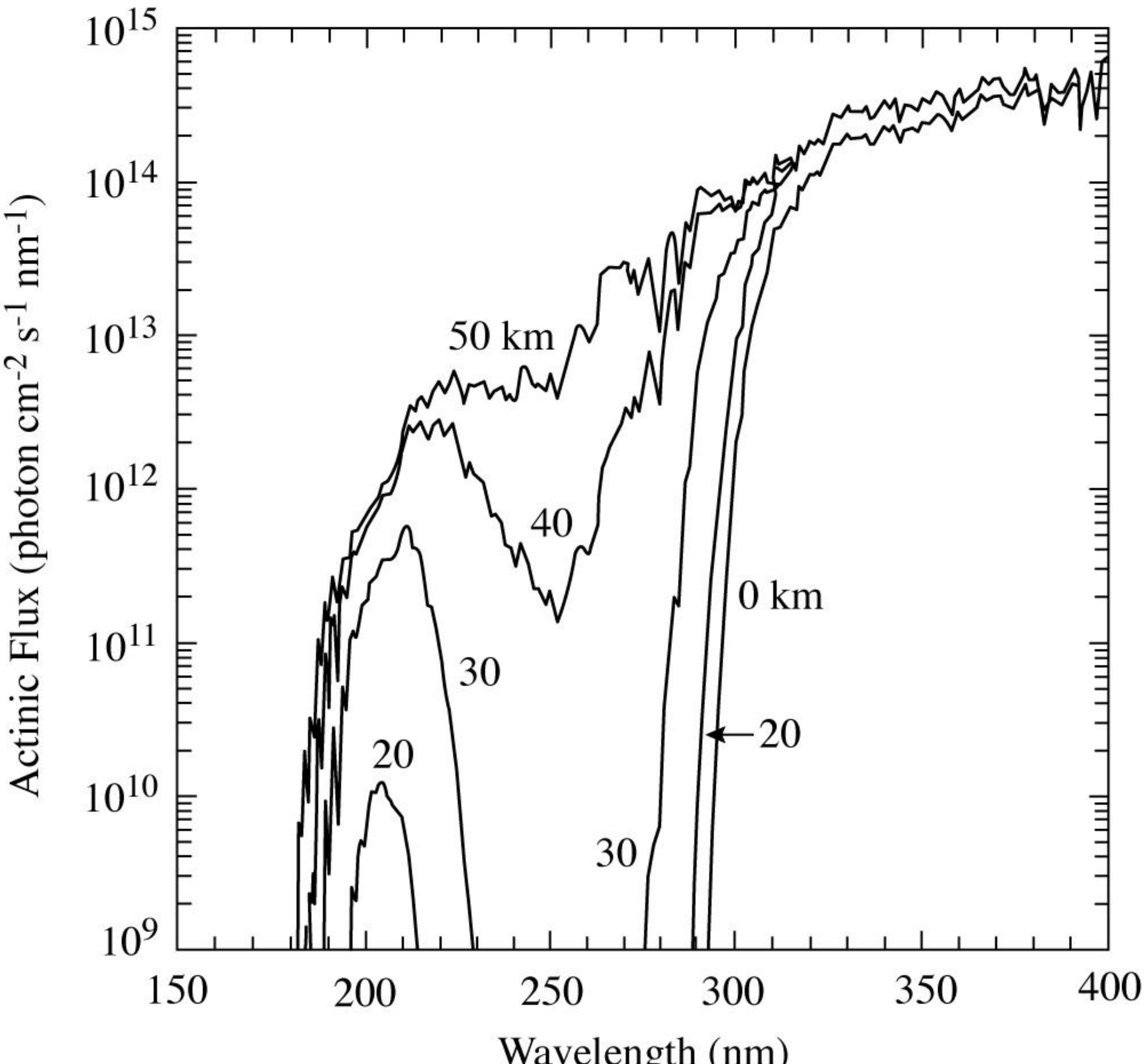
= [R2-R3-R4] + [2R1 + R3 – R2 – R4] = 2 R1 – 2R4 = 0

$$2R1 = 2R4 \Rightarrow k_1[\mathrm{O}_2] = k_4[\mathrm{O}_3][\mathrm{O}] \Rightarrow [\mathrm{O}_3] = \left[\frac{k_1 k_2}{k_3 k_4}\right]^{\frac{1}{2}} C_{\mathrm{O}2} n_a^{\frac{3}{2}}$$

Source: Daniel Jacob

ATMOSPHERIC ATTENUATION OF SOLAR RADIATION

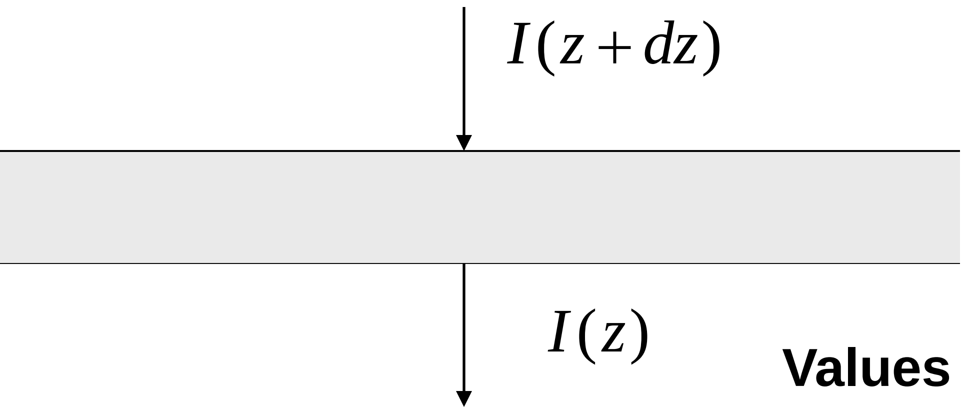
Solar UV radiation reaching the top of the atmosphere is absorbed by ozone



PHOTOLYSIS RATE CONSTANTS: VERTICAL DEPENDENCE

$$\mathrm{X} + h\nu \rightarrow \ldots \qquad\qquad k = \int_0^\infty q_X(\lambda)\sigma_X(\lambda) I_\lambda d\lambda$$

quantum yield | absorption X-section | photon flux

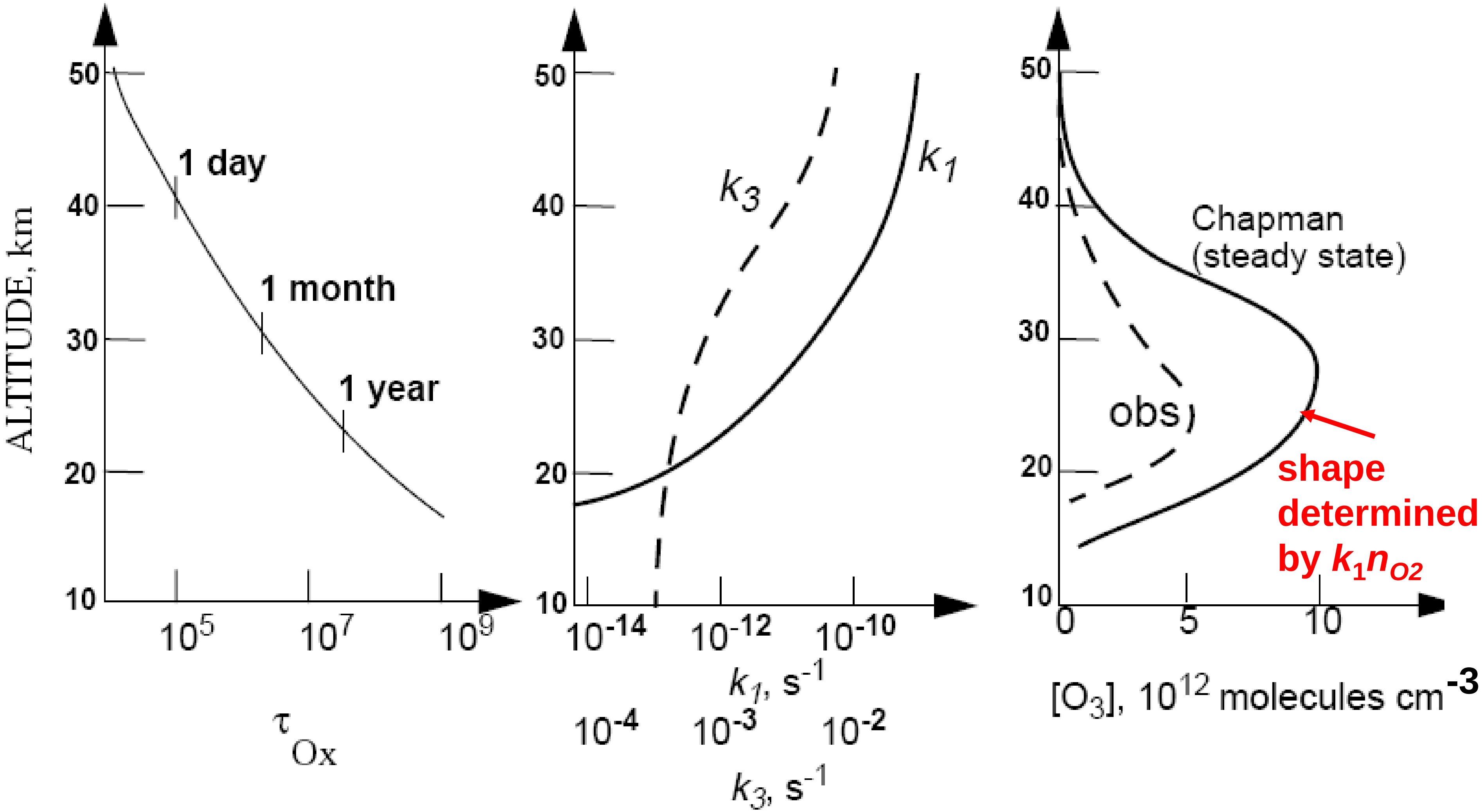


Values of I_λ are less at lower altitudes, so k_1 & k_3 decrease correspondingly

$$I(z) = I(\infty)\mathrm{e}^{-\delta}$$

$$\delta = \int_z^\infty (\sigma_{O2} n_{O2}(z') + \sigma_{O3} n_{O3}(z')) dz'$$

CHAPMAN MECHANISM vs. OBSERVATION



Chapman mechanism reproduces shape, but is too high by factor 2-3

⇨ missing sink!

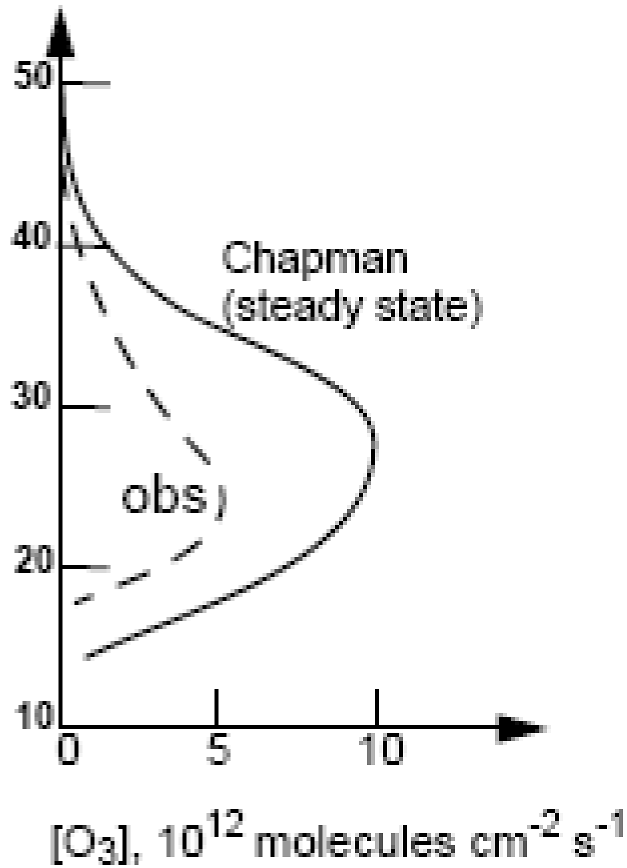
Not Very Straightforward

$$[O_3] = \left(\frac{k_1 k_2}{k_3 k_4}\right)^{\frac{1}{2}} C_{O2} n_a^{\frac{3}{2}}$$

The simplicity of the above expression is misleading, calculating $[O_3]$ from this equation is not straightforward because the photolysis rate constants *$k_1(z)$* & *$k_3(z)$* at altitude *z* depend on the local actinic flux *$I_\lambda(z)$*, which is attenuated due to absorption of radiation by the O_2 & O_3 columns overhead

Thus the above expression is not explicit and R.H.S has some dependency on $[O_3]$ as well !
Then how to solve the problem?
Solve above numerically by starting from the top of the atmosphere with $I_\lambda = I_{\lambda,\infty}$ and $[O_3] = 0$, and progressing downward by small increments in the atmosphere, calculating I_λ (z) and the resulting values of *$k_1(z)$* & *$k_3(z)$* and $[O_3]$ (z) as one penetrates in the atmosphere

RESULTING SOLUTION OF CHAPMAN MECHANISM vs. OZONE OBSERVATIONS



The product $k_1 n_{O2}$ has a maximum at 20-30 km altitude

k_1 ⇧ & $nO2$ ⇩

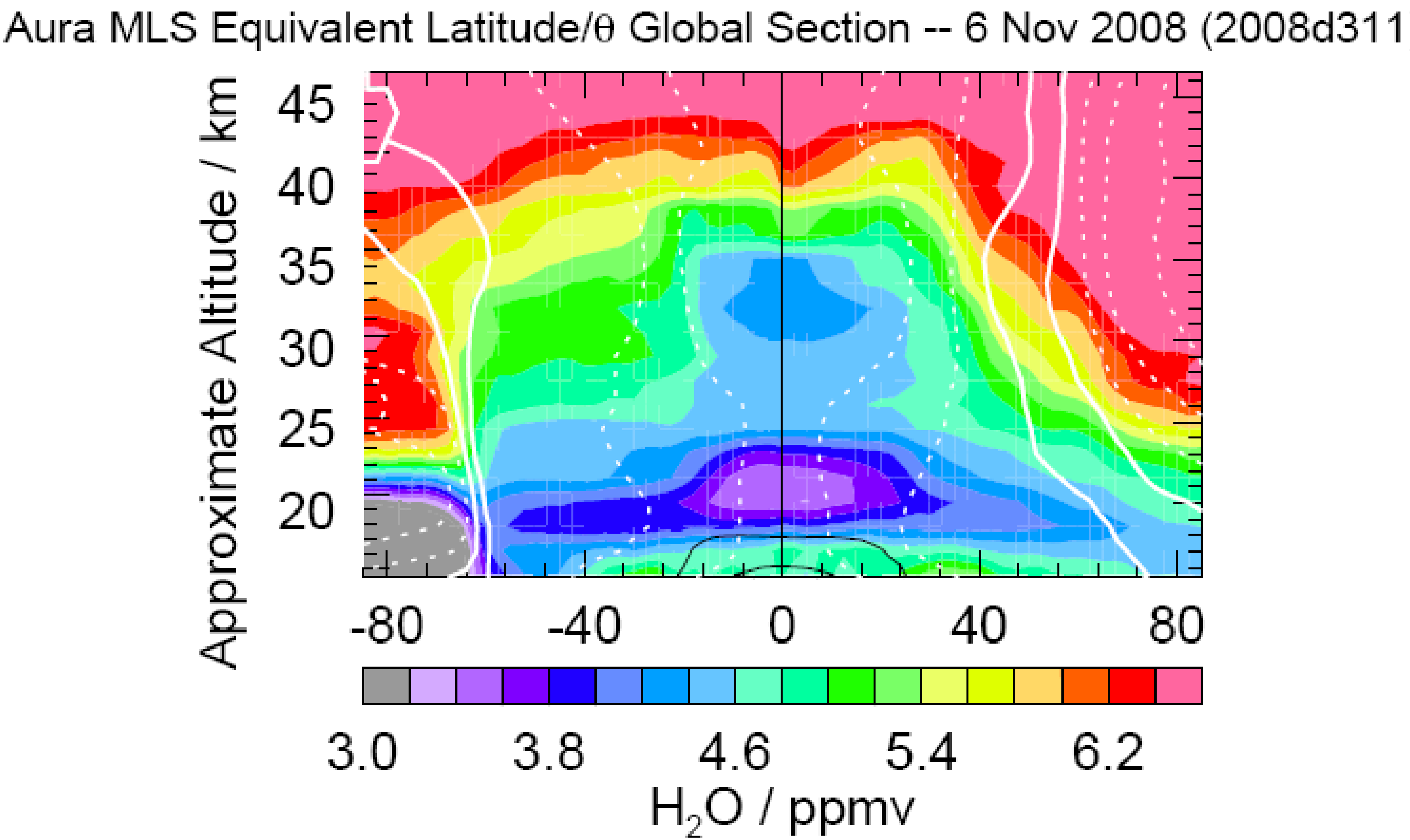
with height

shape determined by $k_1 n_{O2}$

Chapman mechanism reproduces shape, but is too high by factor 2-3

⇨ missing sink!

WATER VAPOR IN STRATOSPHERE



Source: transport from troposphere, oxidation of methane (CH_4)

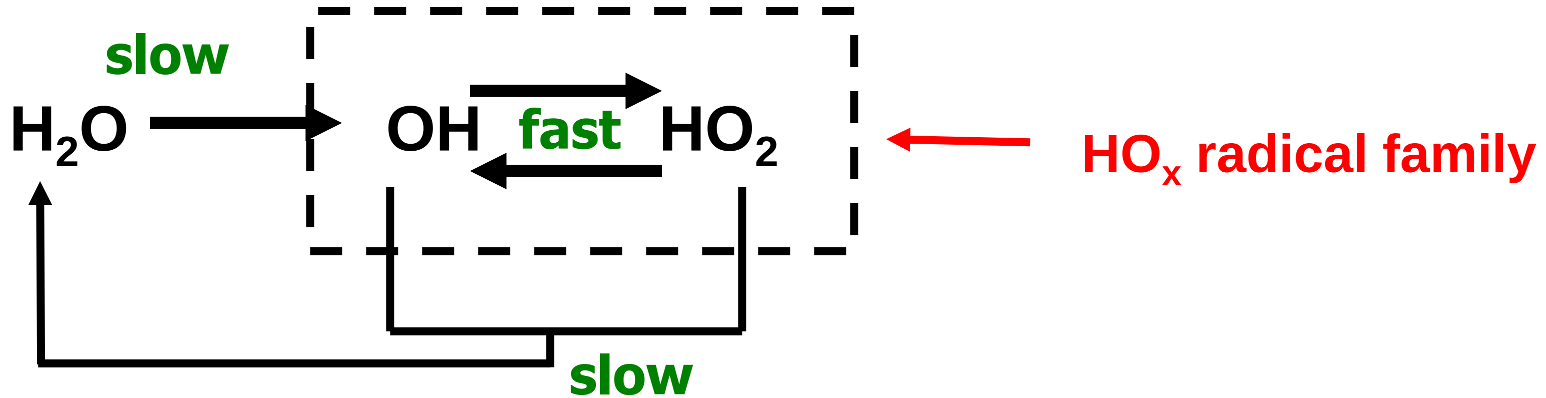
Ozone loss catalyzed by hydrogen oxide ($HO_x \equiv H + OH + HO_2$) radicals

Initiation: $$H_2O + O(^1D) \rightarrow 2OH$$

Propagation:
$$OH + O_3 \rightarrow HO_2 + O_2$$
$$HO_2 + O_3 \rightarrow OH + 2O_2$$

Net: $$2O_3 \rightarrow 3O_2$$

Termination: $$OH + HO_2 \rightarrow H_2O + O_2$$



CATALYSIS BY NO_x : WORK DONE IN 1960s-1970s NOBEL PRIZE 1995, PAUL CRUTZEN

$O_3 + NO \rightarrow NO_2 + O_2$

$NO_2 + h\nu \rightarrow NO + O$

$O + O_2 + M \rightarrow O_3 + M$

THE ABOVE IS A NULL CYCLE FOR O_3

BUT SOME NO_2 CAN REACT WITH O SINGLET;

$$NO_2 + O \rightarrow NO + O_2$$

$$NO + O_3 \rightarrow NO_2 + O_2$$

$$\text{net:} \quad O_3 + O \rightarrow 2O_2$$

EACH CYCLE CONSUMES TWO O_X MOLECULES

RATE LIMITING STEP IS THIS REACTION BECAUSE NO_2 CAN EITHER PHOTOLYZE OR REACT WITH O

TERMINATION STEP

$$\boxed{NO_2 + OH + M \rightarrow HNO_3 + M}$$

ALSO $N_2O + O \rightarrow 2\ NO$ in 1970s Crutzen et al.

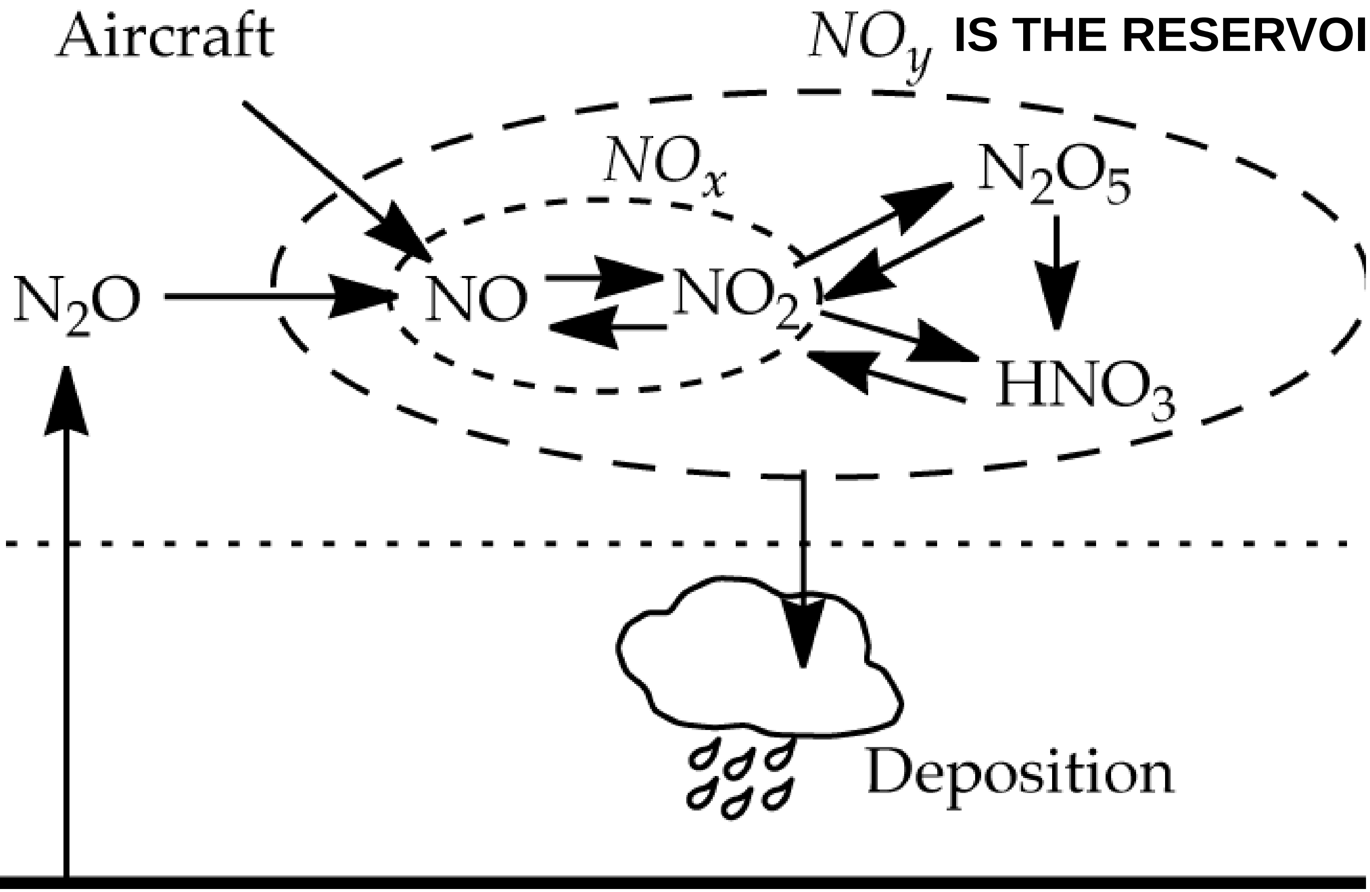
- ✓ **Together with HO_X catalysis closed the budget of Stratospheric ozone**
- ✓ **Along with Johnston's work led to scrapping of supersonic fleet in USA in early 1970s though Concorde in Europe still continued for some time**

ATMOSPHERIC CYCLING OF NO_x AND NO_y

OH IS AVAILABLE ONLY DURING DAY SO AT NIGHT,

$$NO_2 + O_3 \rightarrow NO_3 + O_2$$

$$NO_3 + NO_2 + M \rightarrow N_2O_5 + M$$



AT DAY,
NO_y RELEASES NO_x

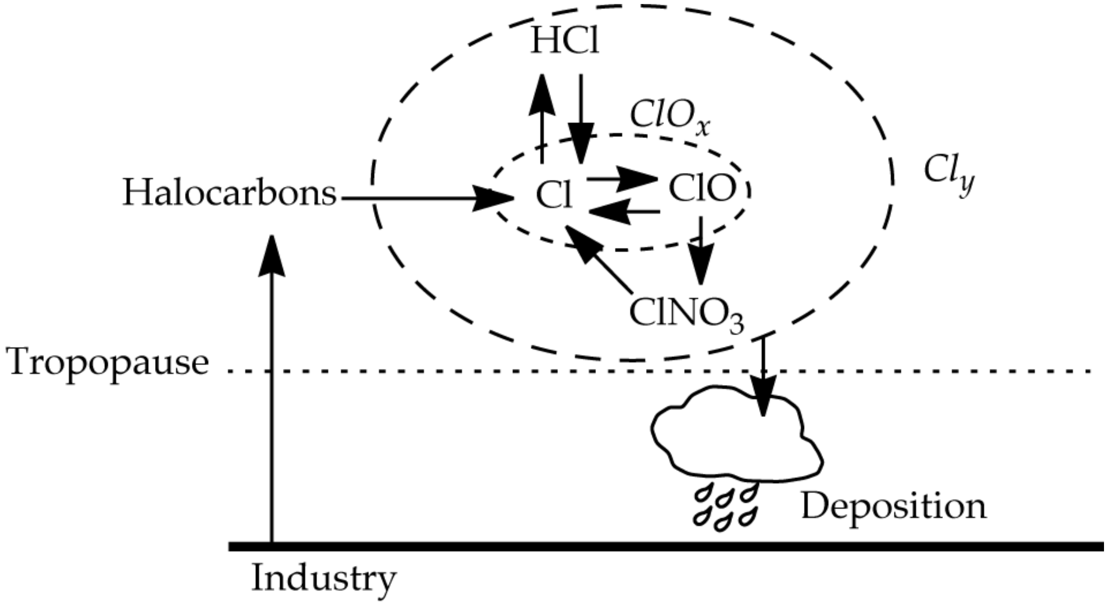
$$HNO_3 + h\nu \rightarrow NO_2 + OH$$

$$HNO_3 + OH \rightarrow NO_3 + H_2O$$

$$NO_3 + h\nu \rightarrow NO_2 + O$$

$$N_2O_5 + h\nu \rightarrow NO_3 + NO_2$$

CHLORINE RADICALS (ClO_X & Cl_y)



IN 1974, MARIO MOLINA & SHERWOOD ROWLAND POINTED OUT THE POTENTIAL FOR OZONE LOSS DUE TO CFCs; CO-WINNERS OF NOBEL PRIZE WITH PAUL CRUTZEN

FOR EXAMPLE, CF_2Cl_2 (TRADE NAME CFC-12); Very long lived; transported to Stratosphere and undergoes photolysis there:

$$CF_2Cl_2 + h\nu \rightarrow CF_2Cl + Cl$$

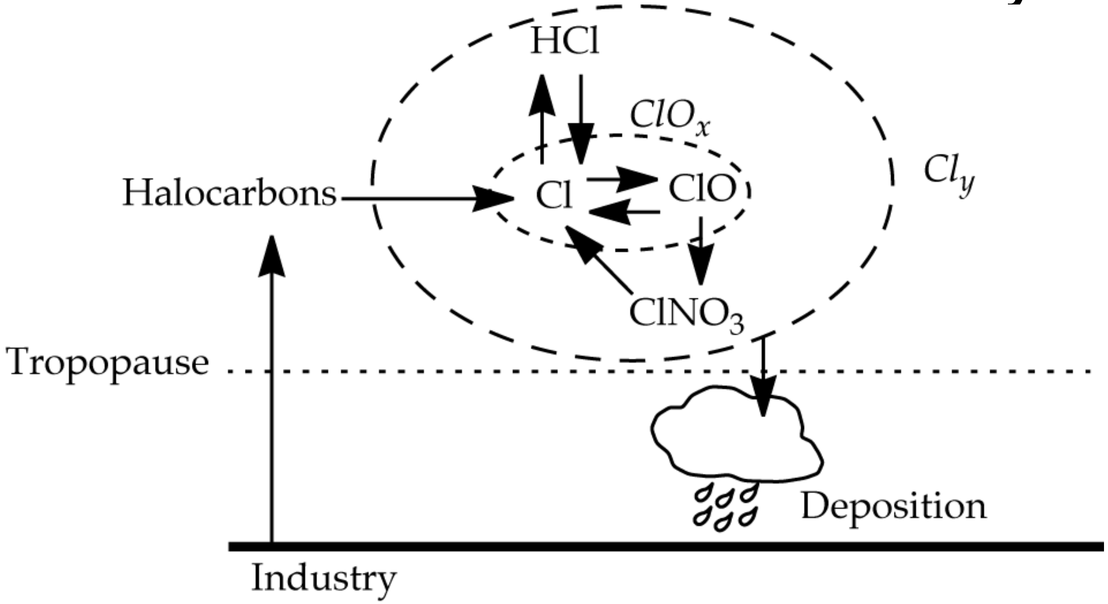
THEN SIMILAR TO NO_X IT CATALYTICALLY DESTROYS OZONE:

$$Cl + O_3 \rightarrow ClO + O_2$$

$$ClO + O \rightarrow Cl + O_2$$ **RATE – LIMITING STEP**

$$\text{net:} \quad O_3 + O \rightarrow 2\,O_2$$

CFCs : SOURCE OF CHLORINE RADICALS (ClO_X & Cl_y)



TERMINATION STEP: CONVERSION OF ClO_X TO NON-RADICAL CHLORINE RESERVOIRS, HCl & $ClNO_3$

$$Cl + CH_4 \rightarrow HCl + CH_3$$

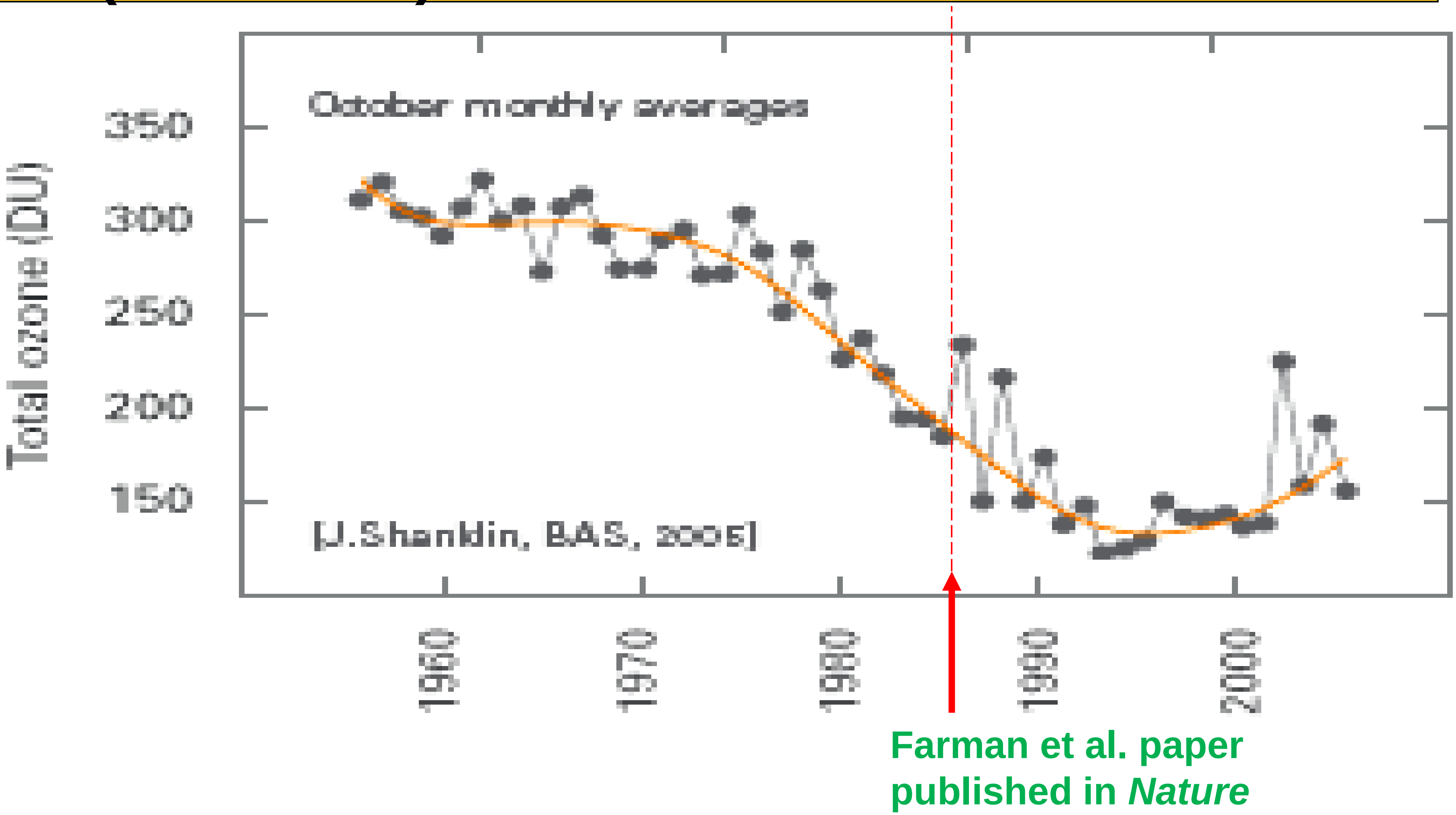
$$ClO + NO_2 + M \rightarrow ClNO_3 + M$$

EVENTUALLY THOUGH,

$$HCl + OH \rightarrow Cl + H_2O$$

$$ClNO_3 + h\nu \rightarrow Cl + NO_3$$

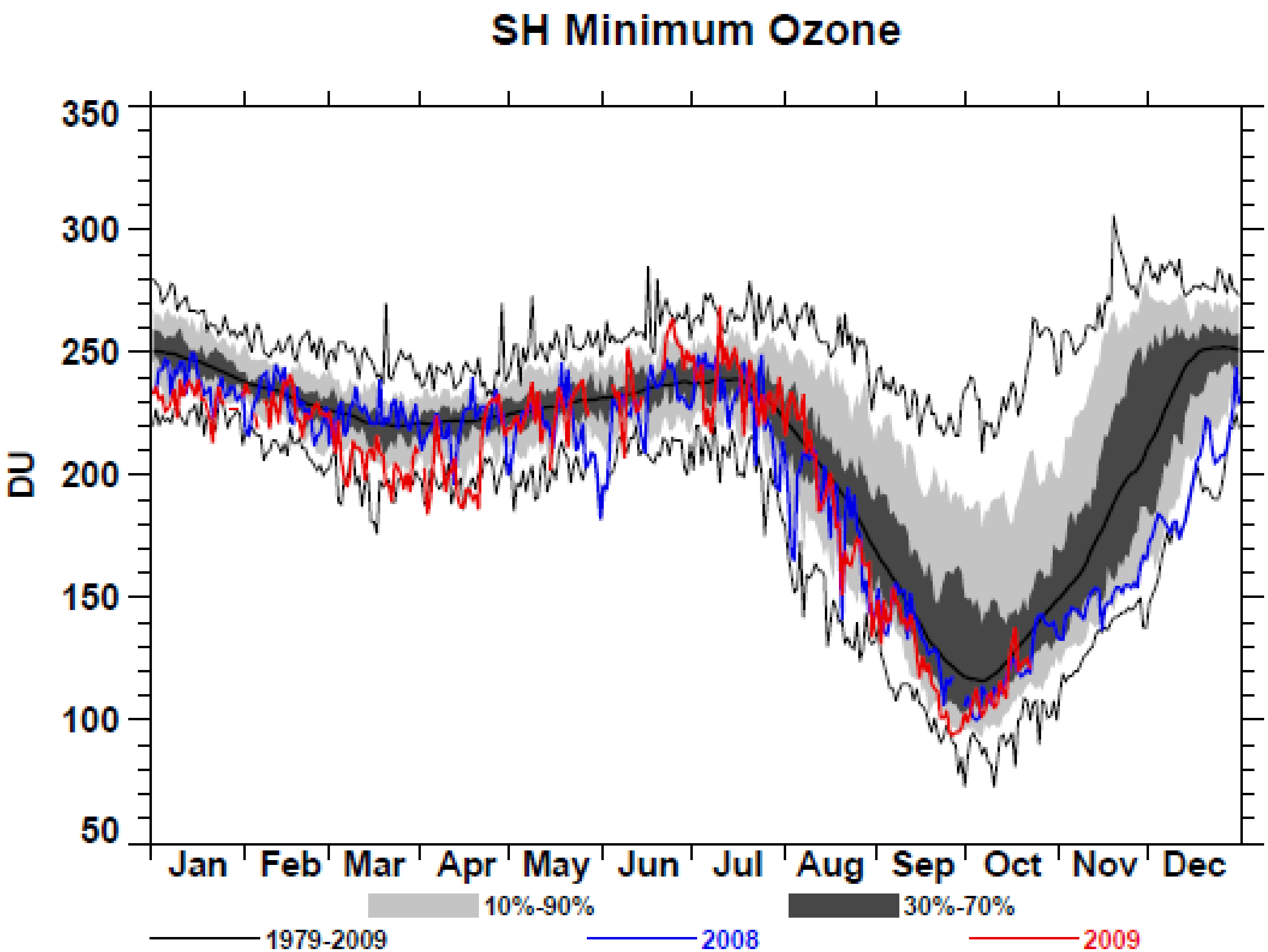
OZONE TREND AT HALLEY BAY, ANTARCTICA (OCTOBER):DISCOVERY OF OZONE HOLE



1 Dobson Unit (DU) = 0.01 mm O_3 STP = 2.69×10^{16} molecules cm^{-2}

SLIDE: DANIEL JACOB

THE OZONE HOLE IS A SPRINGTIME PHENOMENON

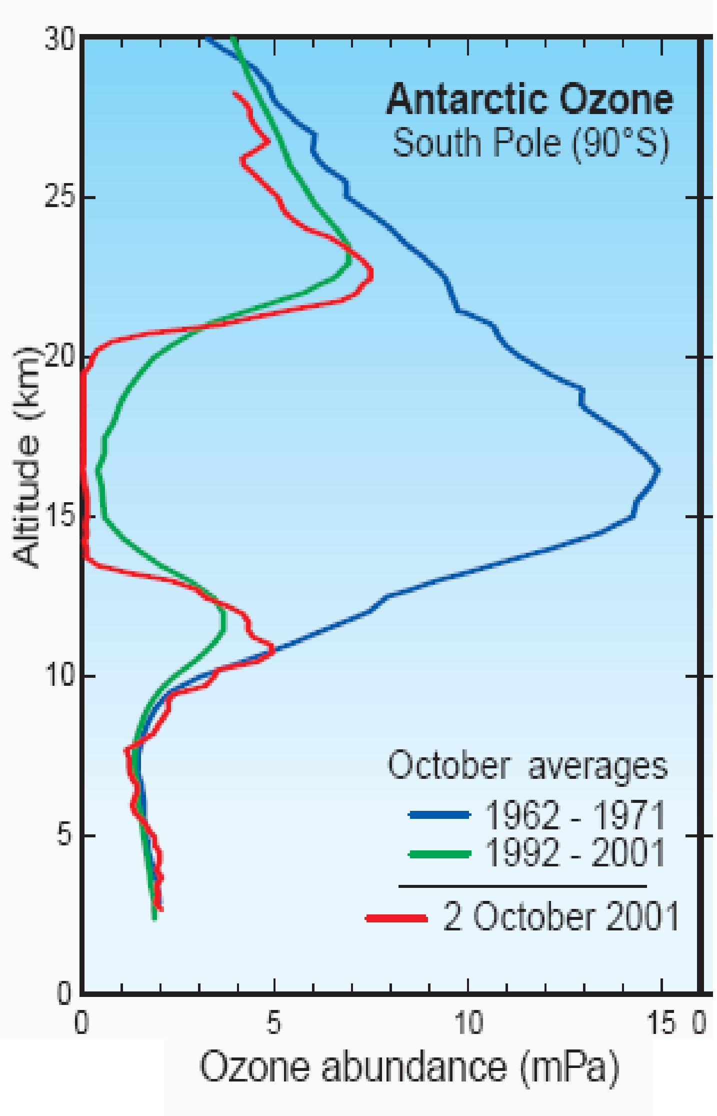


SLIDE: DANIEL JACOB

VERTICAL STRUCTURE OF THE OZONE HOLE: near-total depletion in lower stratosphere

Argentine Antarctic station

Southern tip of S. America



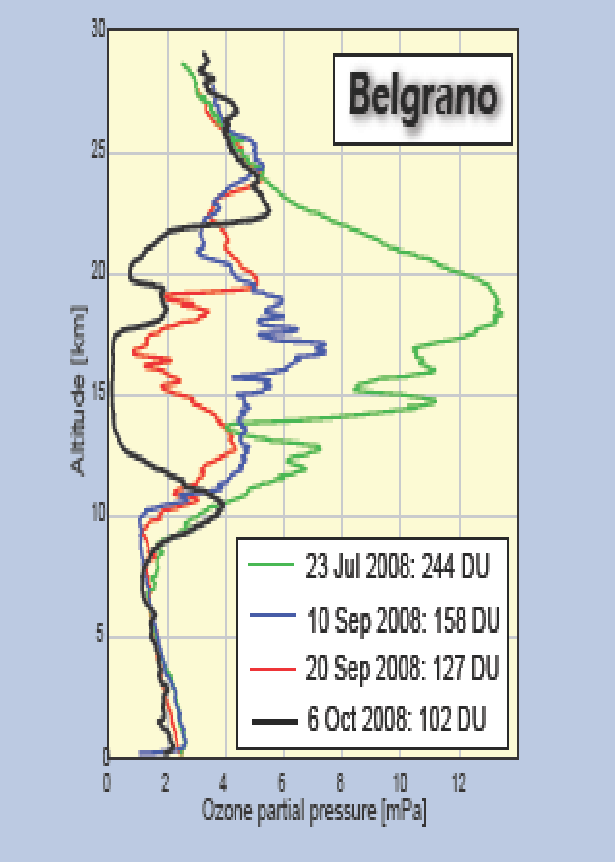
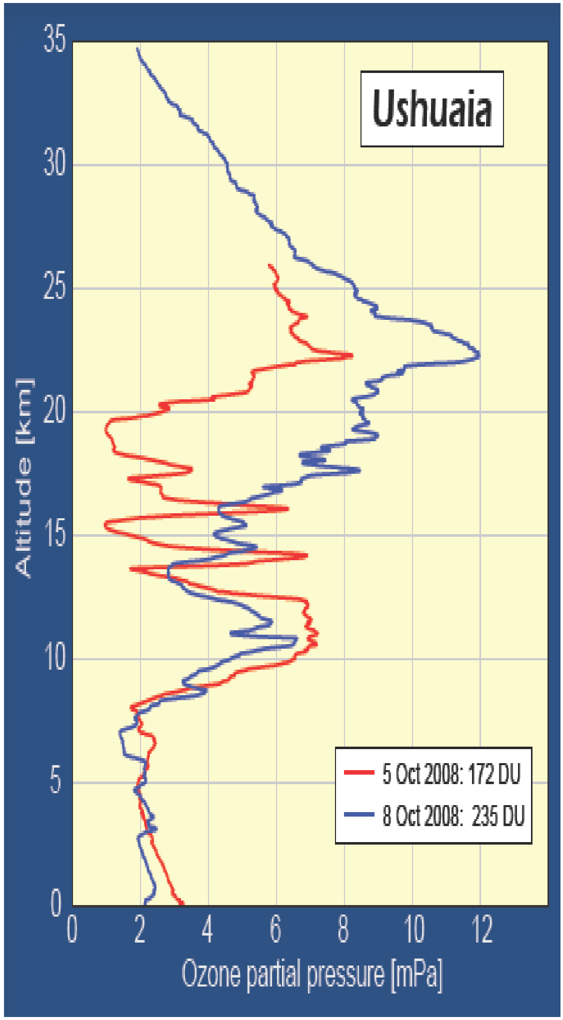


Figure 10. Ozonesonde profiles measured at the Argentine GAW station Belgrano (77.9°S, 34.6°W) on 23 July (green), 10 September (blue) and 20 September 2008.



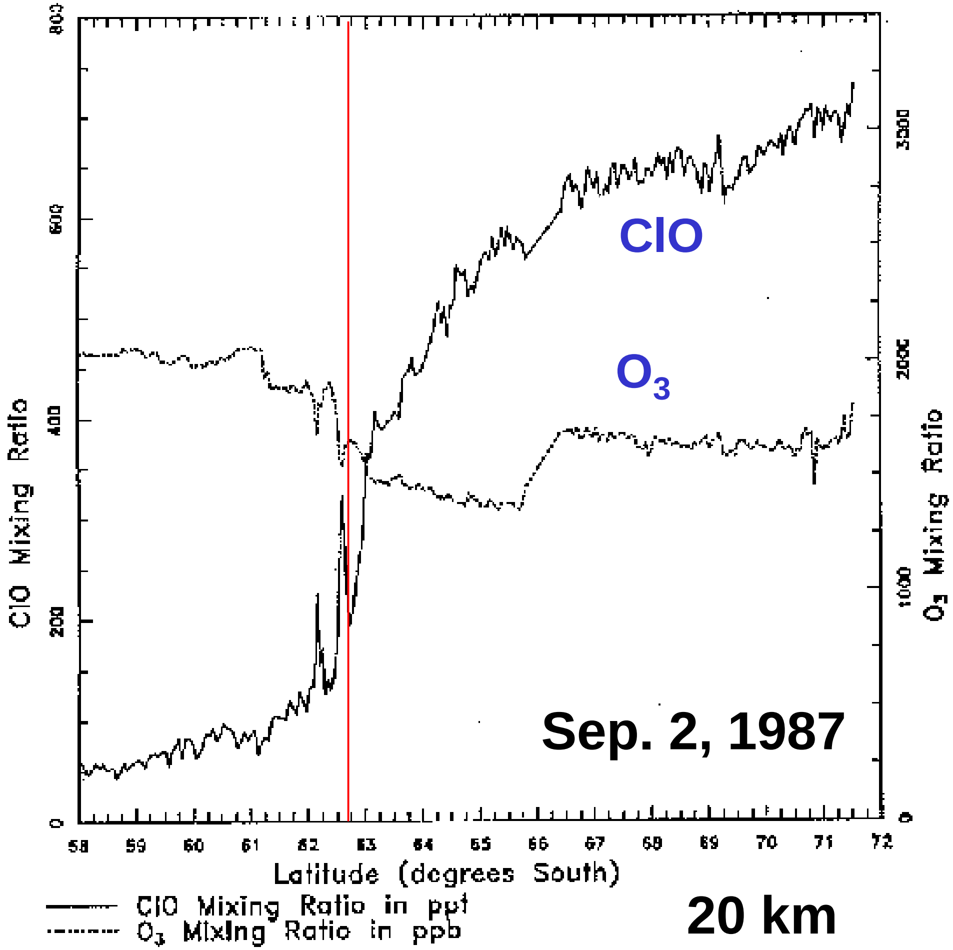
A MYSTERY IN THE CHEMISTRY

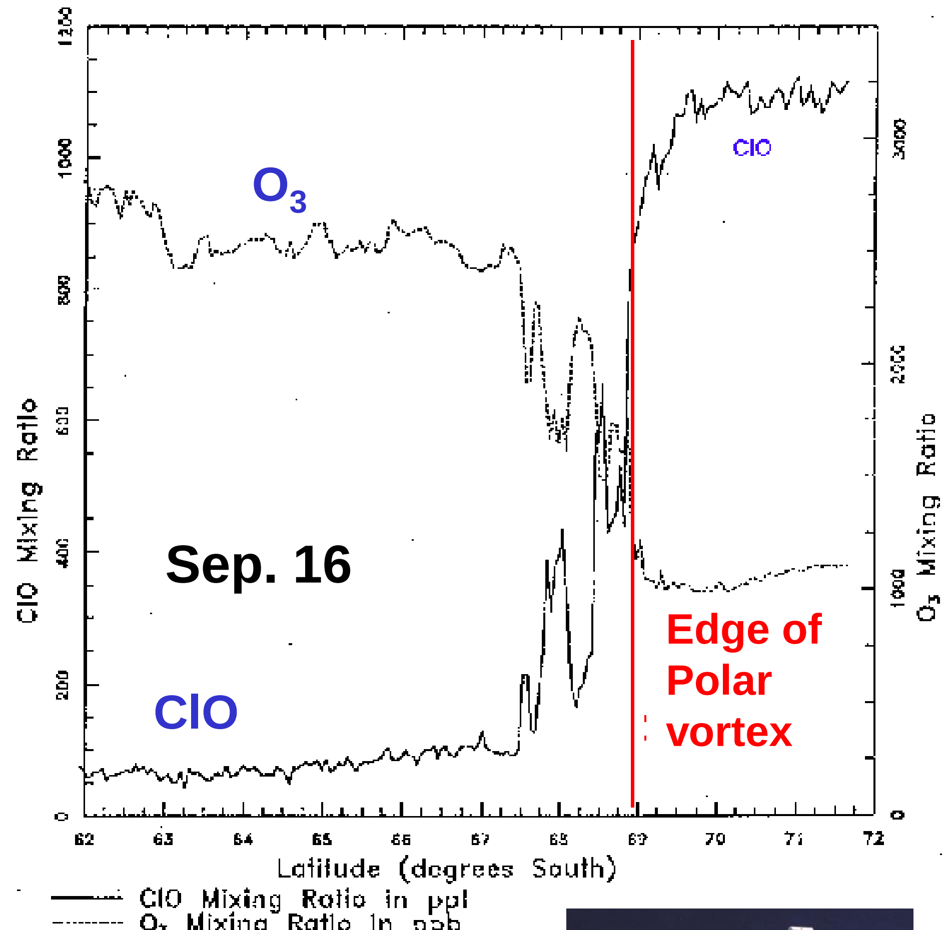
Reaction with O (^{1}D) of NO_2 & ClO could not explain this springtime ozone loss as [O] is very low under the low light conditions prevalent in the Antarctic

Studies found depletion of Ozone to be associated with Exceptionally high ClO, a result confirmed by satellite data

ASSOCIATION OF ANTARCTIC OZONE HOLE WITH HIGH LEVELS OF ClO

Sept. 1987 ER-2 aircraft measurements at 20 km altitude south of Punta Arenas





20 km altitude

Measurements by Jim Anderson's group (Harvard)

A MYSTERY IN THE CHEMISTRY

Studies found depletion of Ozone to be associated with Exceptionally high ClO, a result confirmed by satellite data

Parallel laboratory experiments revealed that at such high ClO levels, a new catalytic cycle involving self reaction of ClO could account for most observed Ozone depletion

$$ClO + ClO + M \rightarrow ClOOCl + M$$
$$ClOOCl + h\nu \rightarrow ClOO + Cl$$
$$ClOO + M \rightarrow Cl + O_2$$
$$(2x) \quad Cl + O_3 \rightarrow ClO + O_2$$

KEY DISCOVERY HERE WAS THAT PHOTOLYSIS OF ClO OCCURS AT O-Cl BOND RATHER THAN O-O BOND WHICH IS WEAKER NOTE IT DOES NOT DEPEND ON [O]

$$\text{net:} \quad 2\,O_3 \rightarrow 3\,O_2$$

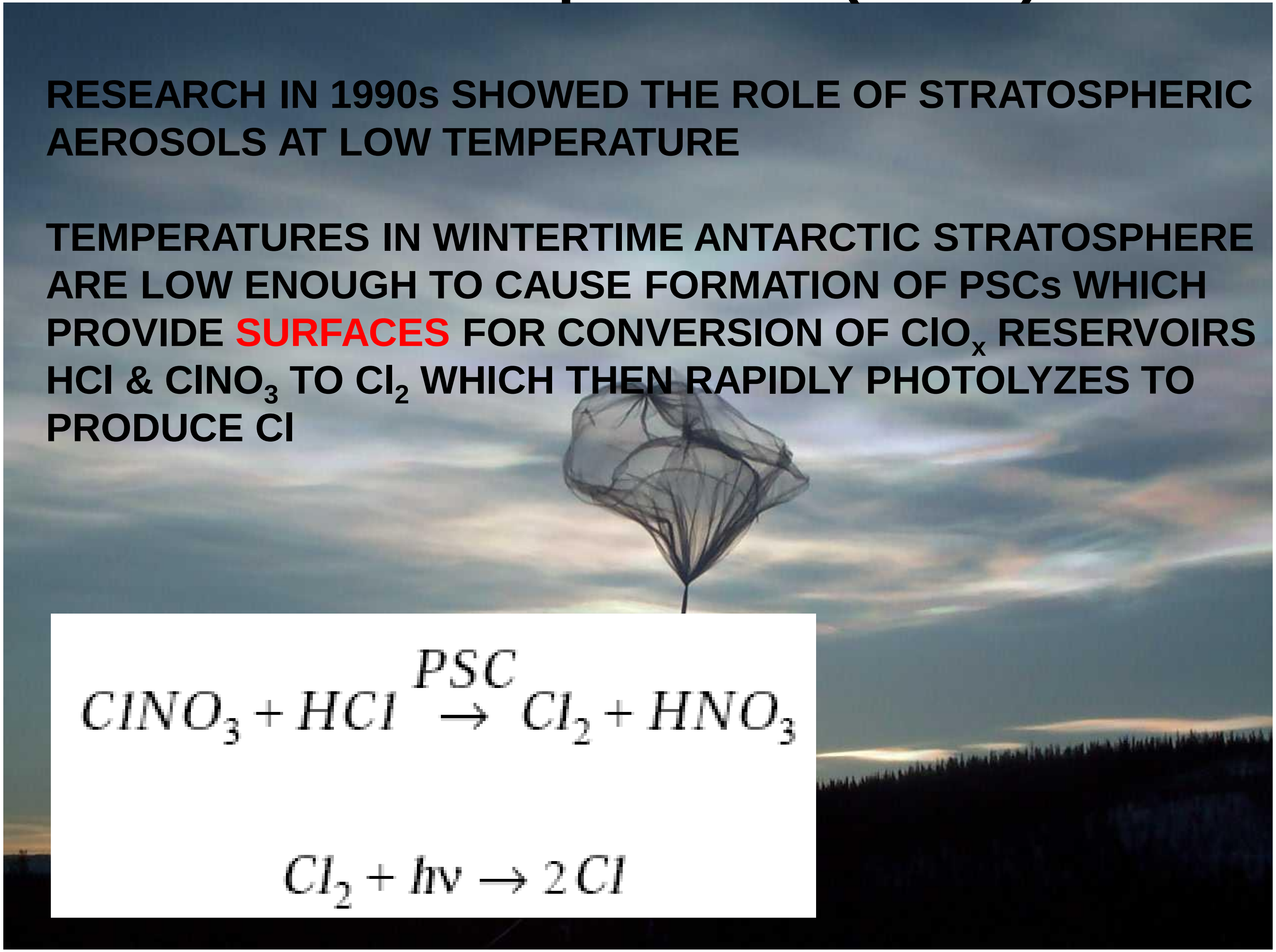
~70% LOSS IN ANTARCTIC OZONE HOLE

AND IF Br IS PRESENT,

$$Br + O_3 \rightarrow BrO + O_2$$
$$BrO + ClO \rightarrow Br + Cl + O_2$$

~ 30% LOSS BY THIS MECHANISM

WHY THE HIGH ClO IN THE ANTARCTIC VORTEX?
Chlorine radicals from reactions of Reservoir species in (PSCs)



RESEARCH IN 1990s SHOWED THE ROLE OF STRATOSPHERIC AEROSOLS AT LOW TEMPERATURE

TEMPERATURES IN WINTERTIME ANTARCTIC STRATOSPHERE ARE LOW ENOUGH TO CAUSE FORMATION OF PSCs WHICH PROVIDE SURFACES FOR CONVERSION OF ClO_x RESERVOIRS HCl & $ClNO_3$ TO Cl_2 WHICH THEN RAPIDLY PHOTOLYZES TO PRODUCE Cl

$$ClNO_3 + HCl \xrightarrow{PSC} Cl_2 + HNO_3$$

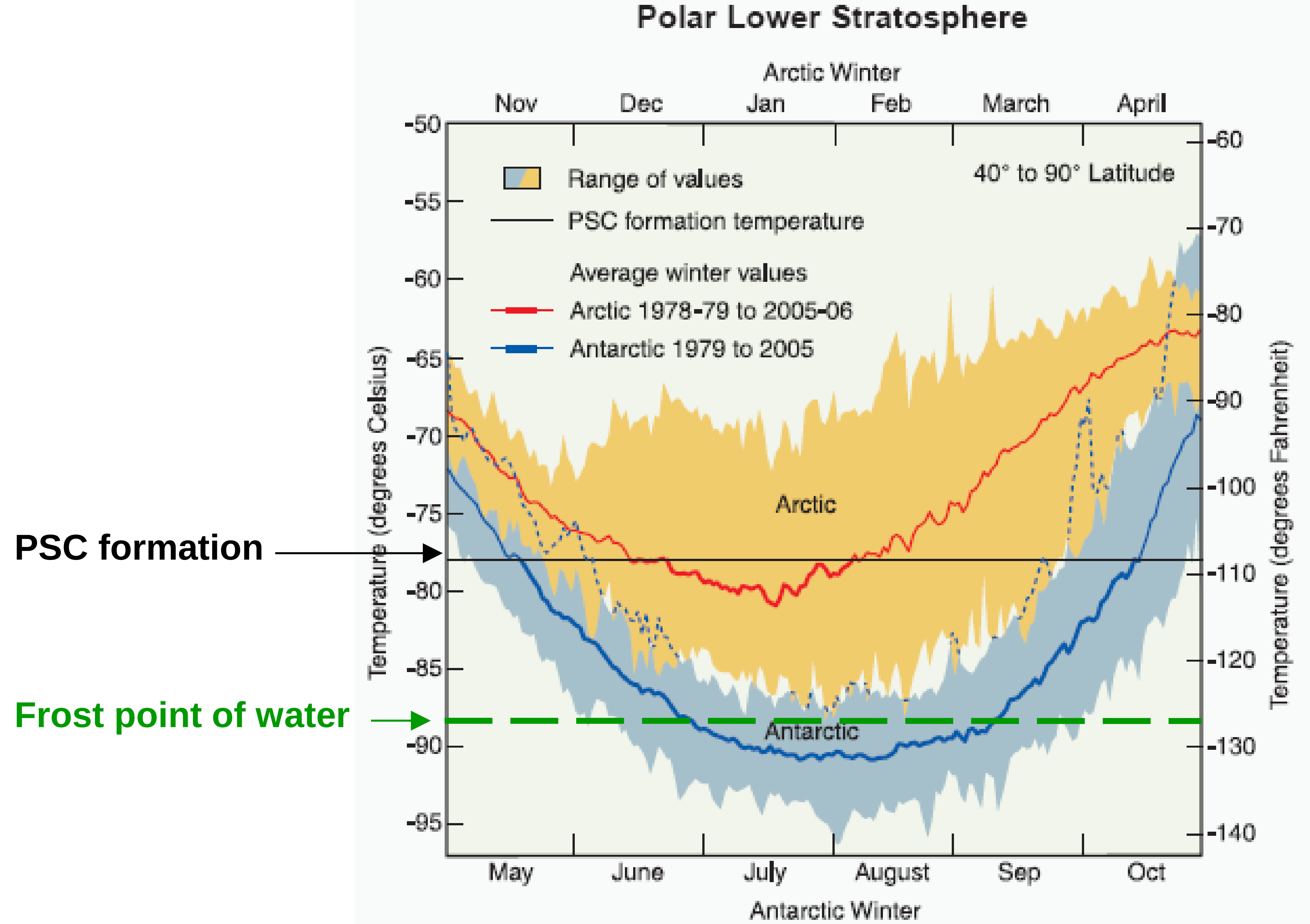
$$Cl_2 + h\nu \rightarrow 2\,Cl$$

WHY ARE PSCs FORMED IN THE ANTARCTIC?

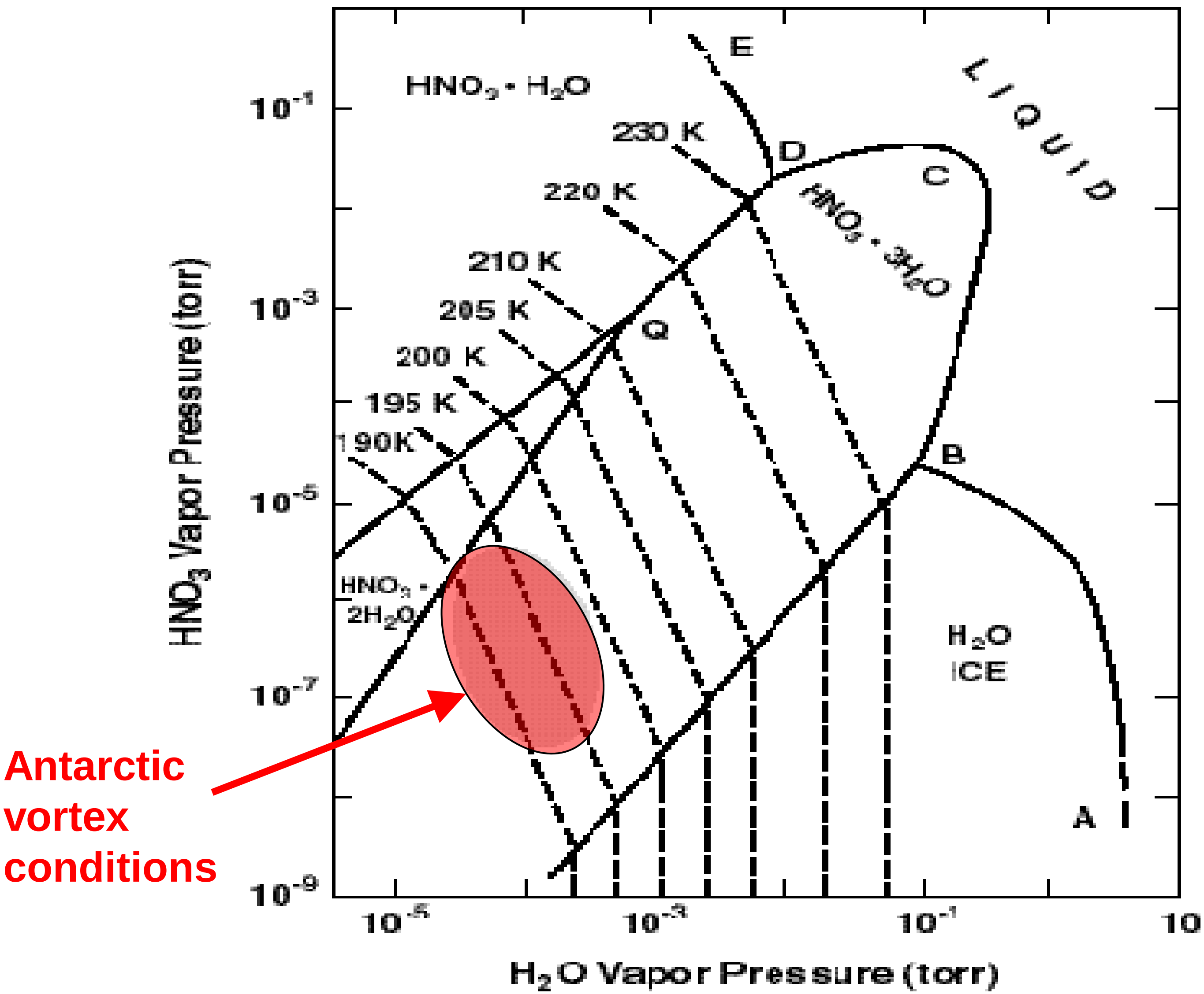
• Stratosphere is extremely dry and so condensation of water vapour to form ice clouds requires very low temperatures
• 3-5 ppmV corresponds to water vapour pressure of 3-5 x 10^{-4} hPa at which the frost point is 185-190 K & such low temperatures rarely occur (only during antarctic winter)
• When temperatures do fall below 190 K, PSC's are observed which form already at 197 K, and this higher threshold for PSC formation is responsible for the large extent of the O_3 hole
• Stratosphere also contains enough HNO_3 so that solid HNO_3-H_2O phases may form, namely, NAT, NAD & NAM

LACK OF TOPOGRAPHY & DIFF. LAND SEA HEATING LEADS TO VERY LITTLE MERIDIONAL TRANSPORT & EFF. ISOLATION OF THE ANATRCTIC WHICH IS STRONGEST IN WINTER

PSC FORMATION AT COLD TEMPERATURES



HOW DO PSCs START FORMING AT 195K?
HNO_3-H_2O PHASE DIAGRAM

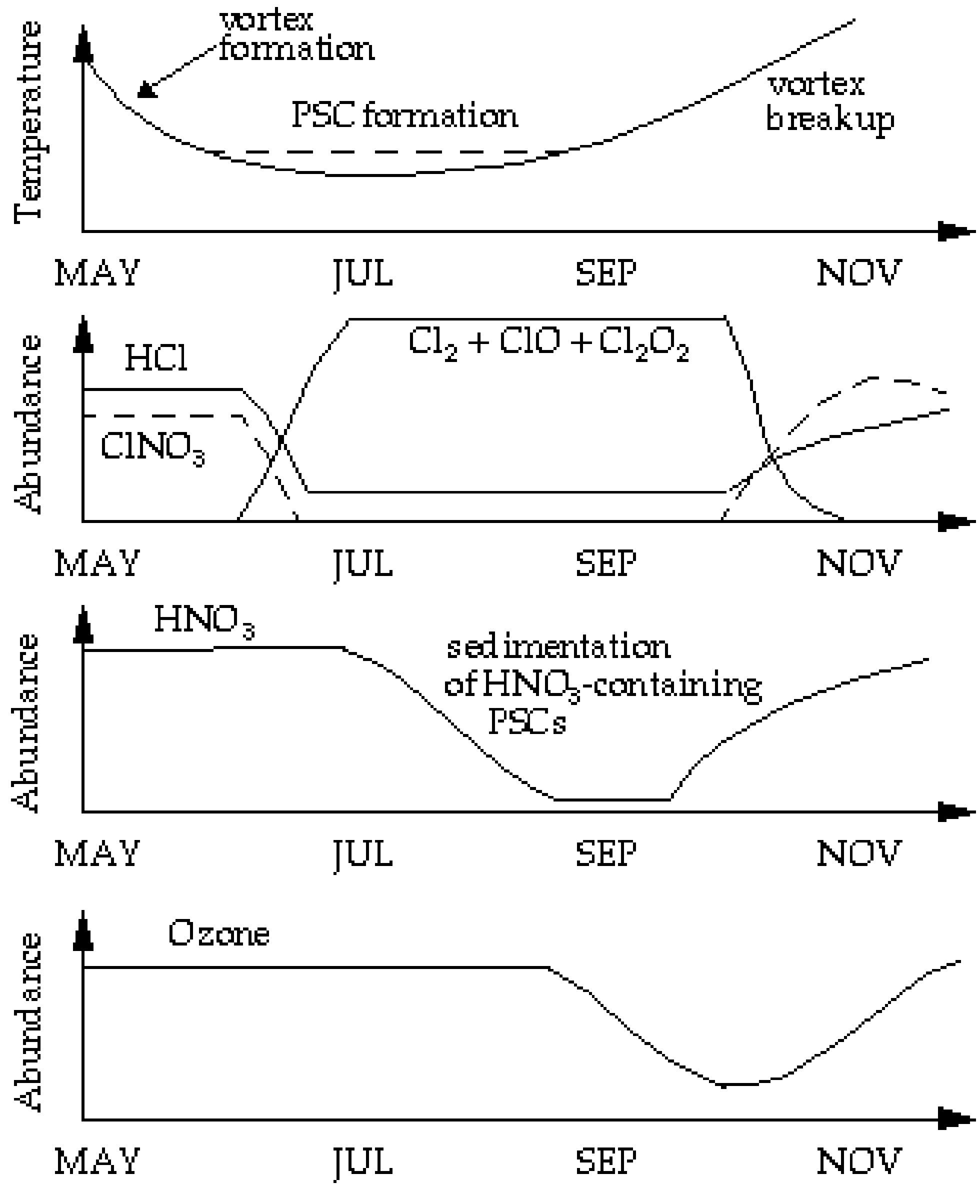


6 phases
H20 ice
Liq. HNO3-H20
Gas HNO3-H20
NAT
NAD
NAM

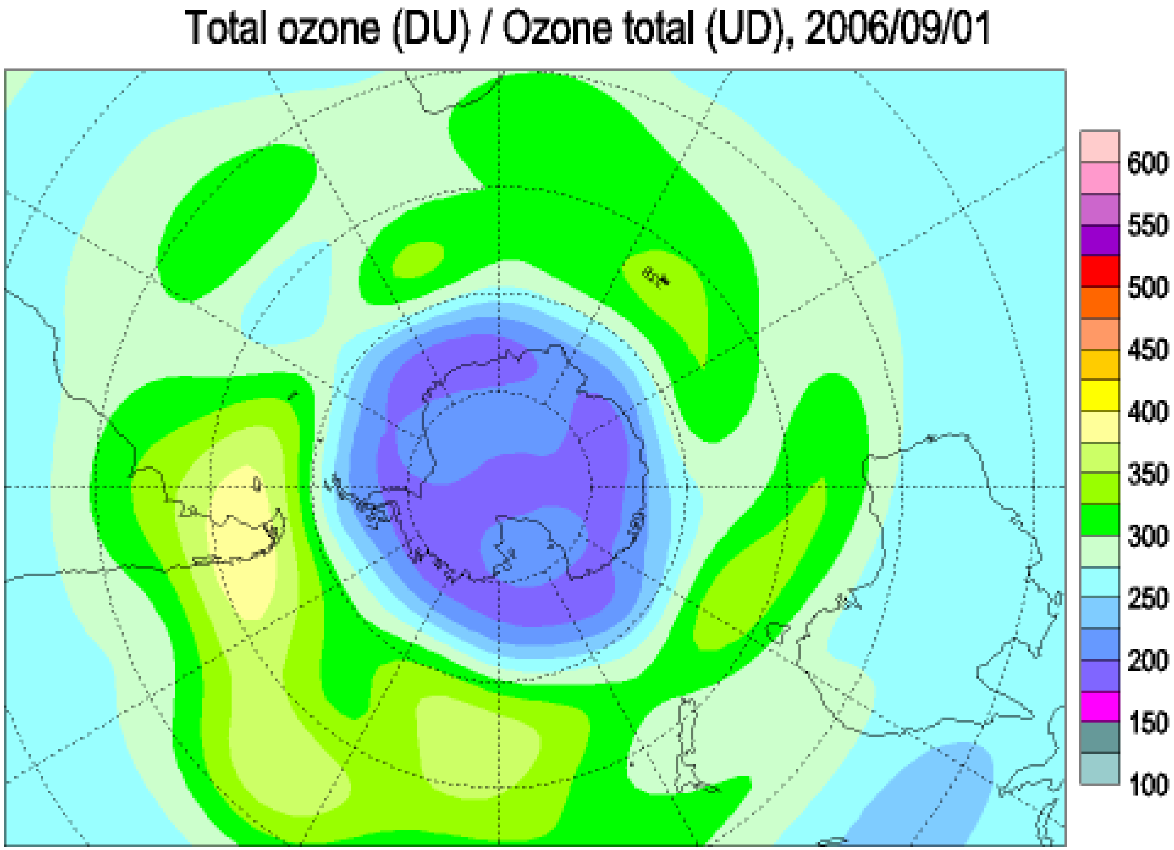
Number of n Independent variables defining the T.D stable system is
n= c+p -2
Eqb of PSC with Gas phase
= 2+2-2
= 2, so given HNO3 (g) ,H20(g) we can infer fav. phase

PSCs are not water but nitric acid trihydrate (NAT) clouds

CHRONOLOGY OF ANTARCTIC OZONE HOLE

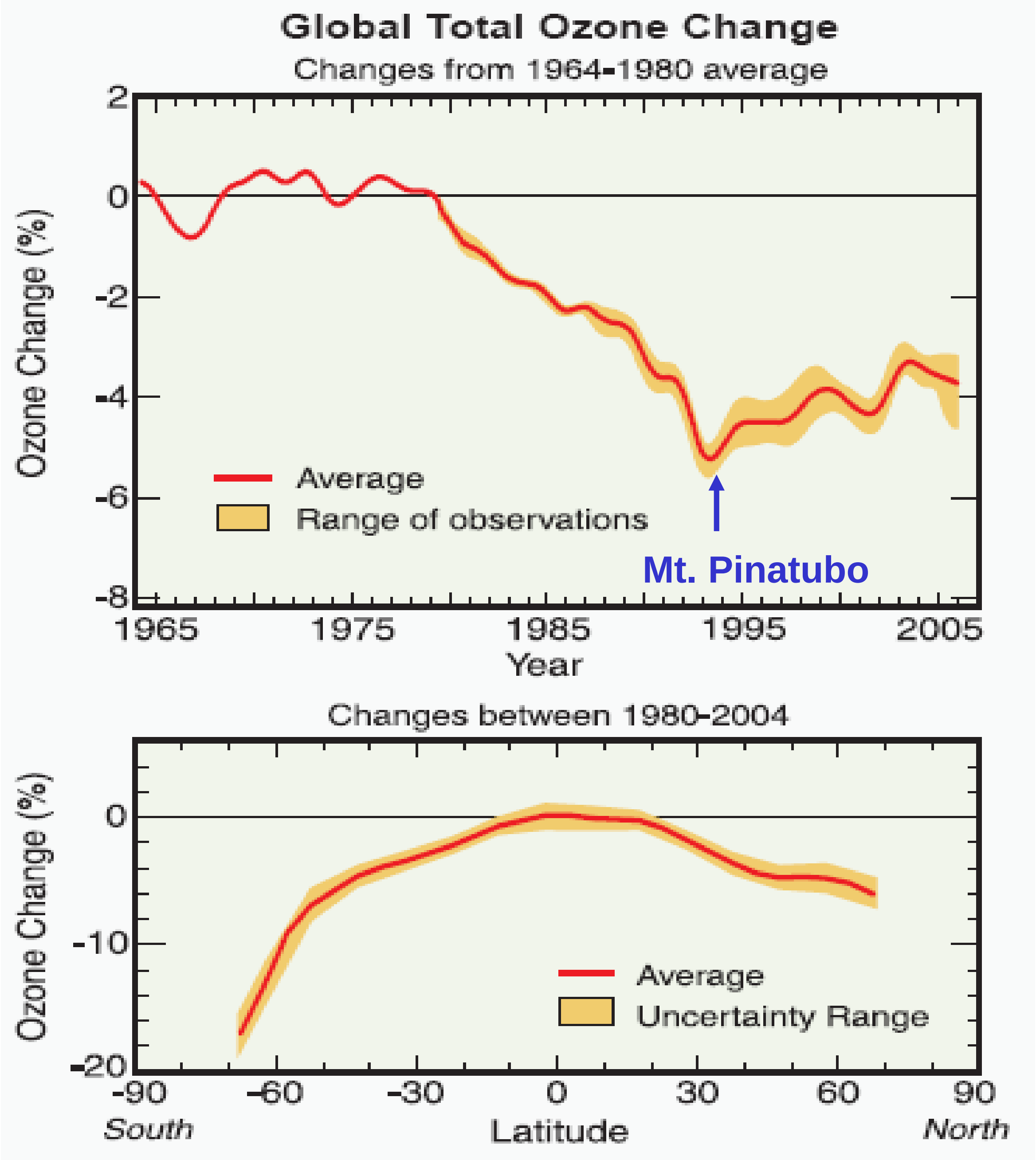


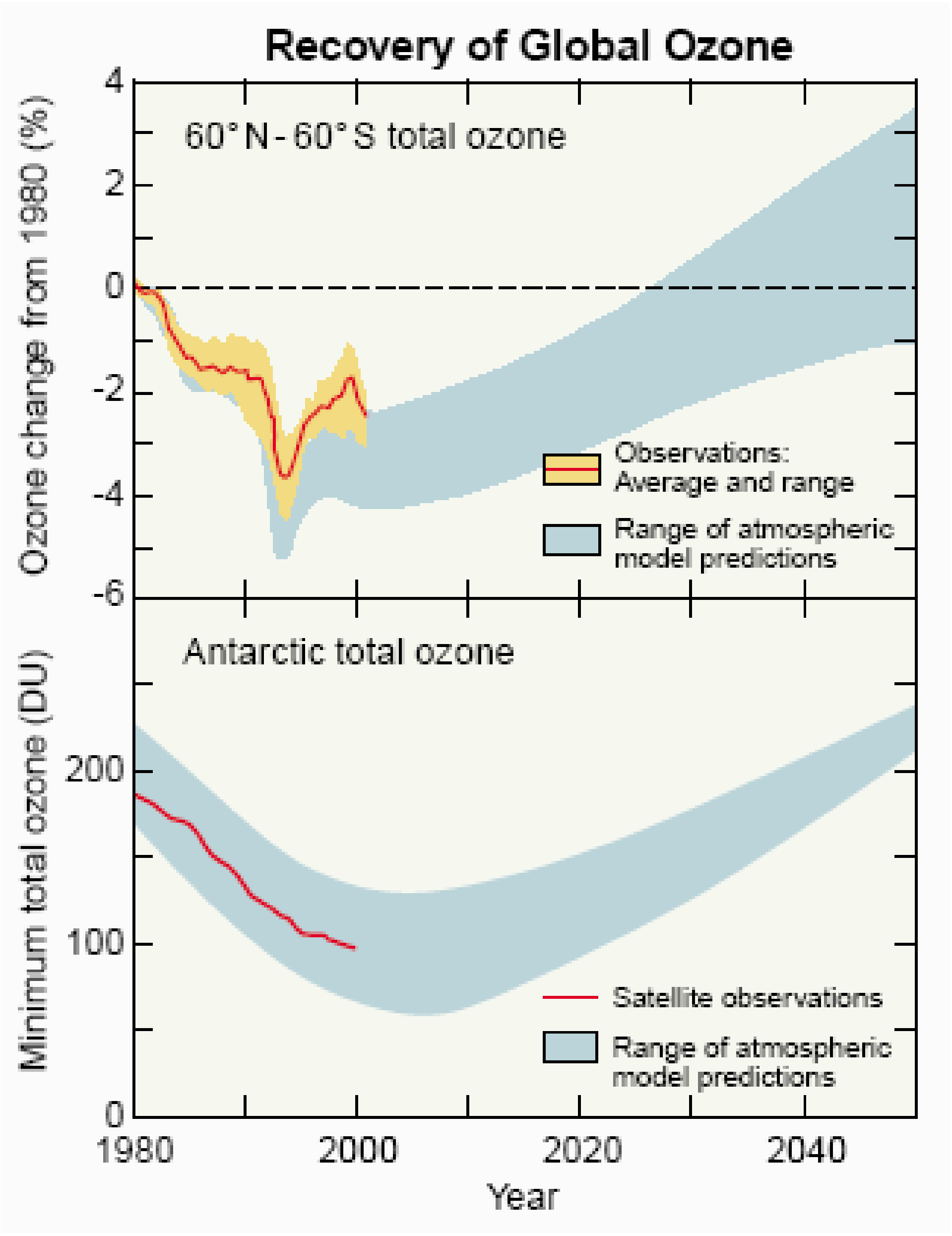
THE POLAR VORTEX (Sep-Oct 2006)



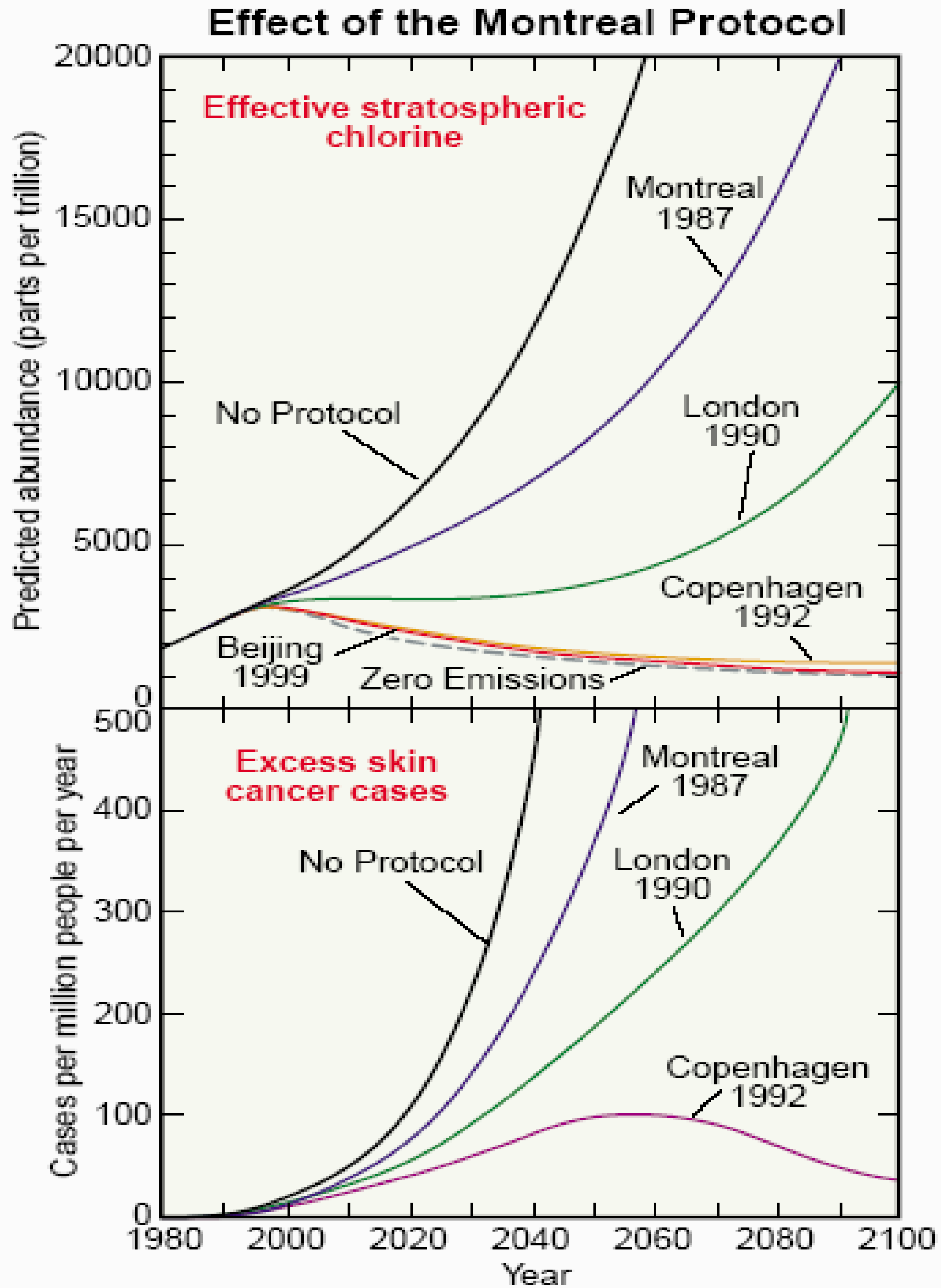
SLIDE: DANIEL JACOB

TRENDS IN GLOBAL OZONE




Recovery of Global Ozone
60° N - 60° S total ozone
Ozone change from 1980 (%)
4
2
0
-2
-4
-6
Observations:
Average and range
Range of atmospheric
model predictions
Antarctic total ozone
Minimum total ozone (DU)
200
100
0
Satellite observations
Range of atmospheric
model predictions
1980
2000
2020
2040
Year

SKIN CANCER EPIDEMIOLOGY PREDICTIONS



THE MONTREAL PROTOCOL: 1987

MOLINA & ROWLAND WARNED THAT ClO_x CATALYZED O_3 LOSS WOULD DAMAGE THE OZONE LAYER IF EMISSIONS OF CFCs DID NOT STOP. IN 1985 OBSERVATIONS OF OZONE HOLE IN ANTARCTICA WERE REPORTED IN NATURE BY FARMAN ET AL. A SERIES OF INTERNATIONAL AGREEMENTS TO BAN CFCs FOLLOWED WHICH FINALLY LED TO THE MONTREAL PROTOCOL IN 1987

The Montreal Protocol is an international treaty designed to protect the ozone layer by phasing out the production of numerous substances believed to be responsible for ozone depletion. The treaty was opened for signature on September 16, 1987, & entered into force on January 1, 1989, followed by a first meeting in Helsinki, May 1989. Since then, it has undergone seven revisions, in 1990 (London), 1991 Nairobi 1992 (Copenhagen), 1993 (Bangkok), 1995 (Vienna), 1997 (Montreal), and 1999 (Beijing). It is believed that if the international agreement is adhered to, the ozone layer is expected to recover by 2050! It has been ratified by 196 states

$CFCl_3$ (CFC-11), CF_2Cl_2 (CFC-12), $C_2F_3Cl_3$ (CFC-113), $C_2F_4Cl_2$(CFC-114) and C_2F_5Cl (CFC-115)

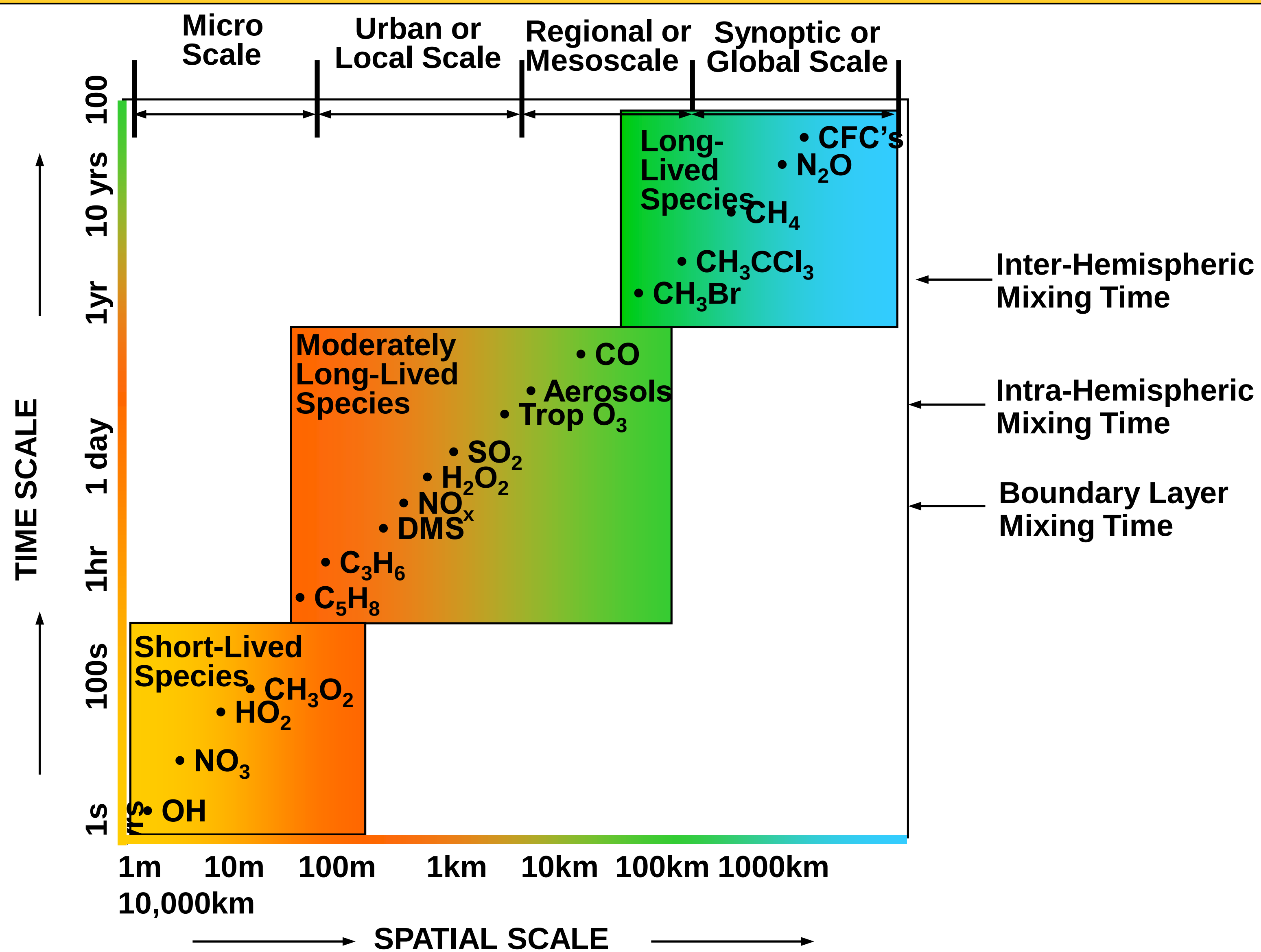
WAS IT EASY TO CONVINCE PEOPLE USING GOOD SCIENCE ?

"The Rowland-Molina hypothesis was strongly disputed by representatives of the aerosol and halocarbon industries. The chair of the board of DuPont was quoted as saying that ozone depletion theory is "a science fiction tale...a load of rubbish...utter nonsense "

- **After publishing their pivotal paper in June 1974, Rowland and Molina testified at a hearing before the U.S. House of Representatives in December 1974. As a result significant funding was made available to study various aspects of the problem and to confirm the initial findings. In 1976, the U.S. National Academy of Sciences (NAS) released a report that confirmed the scientific credibility of the ozone depletion hypothesis**
- **Then, in 1985, British Antarctic Survey scientists Farman, Gardiner and Shanklin shocked the scientific community when they published results of a study showing an ozone "hole" in the journal Nature — showing a decline in polar ozone far larger than anyone had anticipated.**

TROPOSPHERIC CHEMISTRY : CENTRES AROUND OZONE, NO_X , HO_X, VOCs & SECONDARY ORGANIC AEROSOL

Temporal and Spatial Scales of trace compounds



From U. Platt, after Seinfeld and Pandis

Tropospheric Chemistry study triggers: London Smog : 1952, 4000 d

In the 17th century ,John Evelyn's report on air pollution in London from coal read:

It is this horrid Smoke which obscures our Church and makes our Palaces look old, which fouls our Cloth and corrupts the Waters, so as the very Rain, and refreshing Dews which fall in the several Seasons, precipitate to impure vapour, which with its black and tenacious quality, spots and contaminates whatever is exposed to it."

But, without the use of Calculation it is evident to every one who looks on the yearly Bill of Mortality, that near half the children that are born and bred in London die under two years of age. Some have attributed this amazing destruction to luxury and the abuse of Spirituous Liquors: These, no doubts, are powerful assistants; but the constant and unremitting Poison is communicated by the foul Air, which, as the Town still grows larger, has made regular and steady advances in its fatal influence"

Tropospheric Chemistry study triggers: Los Angeles Smog : 1940s

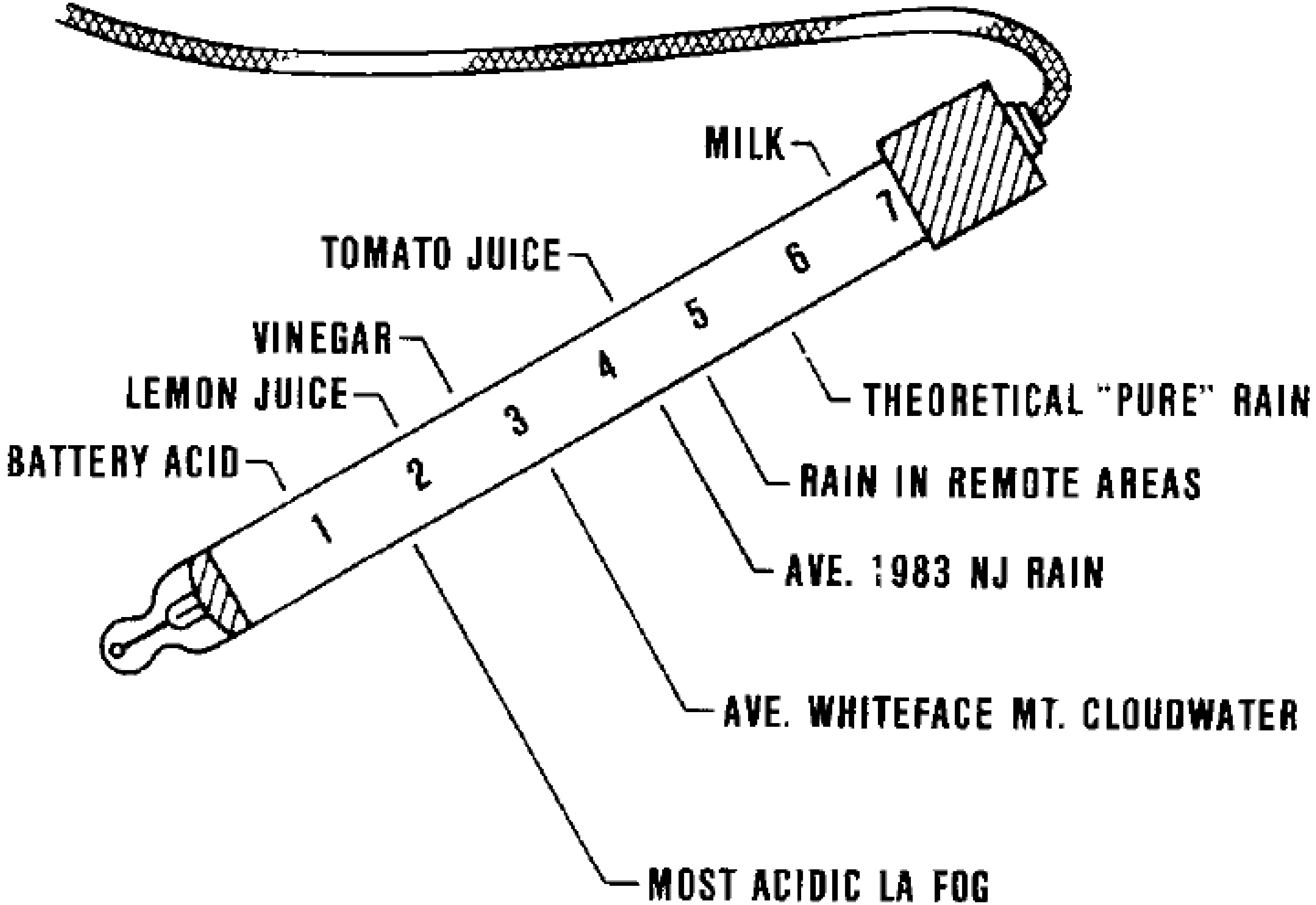
- Observed in LA in the 1940s
  - Hot days
  - Strong sunshine
- Strongly oxidating contaminants, skin, eye-irritations
- Plant damage

- **Arie Haagen Smit 1952: Plant damage reproduced in exp. with**
  - **Organic trace gases, NO_x, sunlight**
  - **Car exhaust, NO_x, sunlight**

VOC + NO_x + sunlight → O_3 + “other products“

Today worldwide problem

Graedel and Crutzen, *Atmospheric Change* (1992)



LONDON SMOG Vs LA SMOG

TABLE 1.2 Historical Aspects of Sulfurous (London) and Photochemical (Los Angeles) Air Pollution

Characteristics	Sulfurous (London)	Photochemical (Los Angeles)
First recognized	Centuries ago	Mid-1940s
Primary pollutants	SO_2, soot particles	VOC, NO_x
Secondary pollutants	H_2SO_4, sulfate aerosols, etc.	O_3, PAN, HNO_3, aldehydes, particulate nitrate and sulfate, etc.
Temperature	Cool (< 35°C)	Hot (> 75°F)
Relative humidity	High, usually foggy	Low, usually hot and dry
Type of inversion	Radiation (ground)	Subsidence (overhead)
Time air pollution peaks	Early morning	Noon to evening

Importance of VOCs in the troposphere

- Formation of tropospheric ozone (O_3) :

$$NO_2 + h\nu \rightarrow NO + O(^3P)$$

$$O(^3P) + O_2 + M \rightarrow O_3 + M$$

$$NO + O_3 \rightarrow NO_2 + O_2$$

Null Cycle Photo-stationary state of O_3 and NO_x

Chain initiation

$$RH + OH \rightarrow R + H_2O$$

$$R + O_2 + M \rightarrow RO_2 + M$$

VOC OH reactivity $= \Sigma k_{(voc_i+OH)}[VOC_i]$

Chain propagation

$$RO_2 + NO \rightarrow RO + \boxed{NO_2}$$

$$RO + O_2 \rightarrow R'O + HO_2$$

$$HO_2 + NO \rightarrow OH + \boxed{NO_2}$$

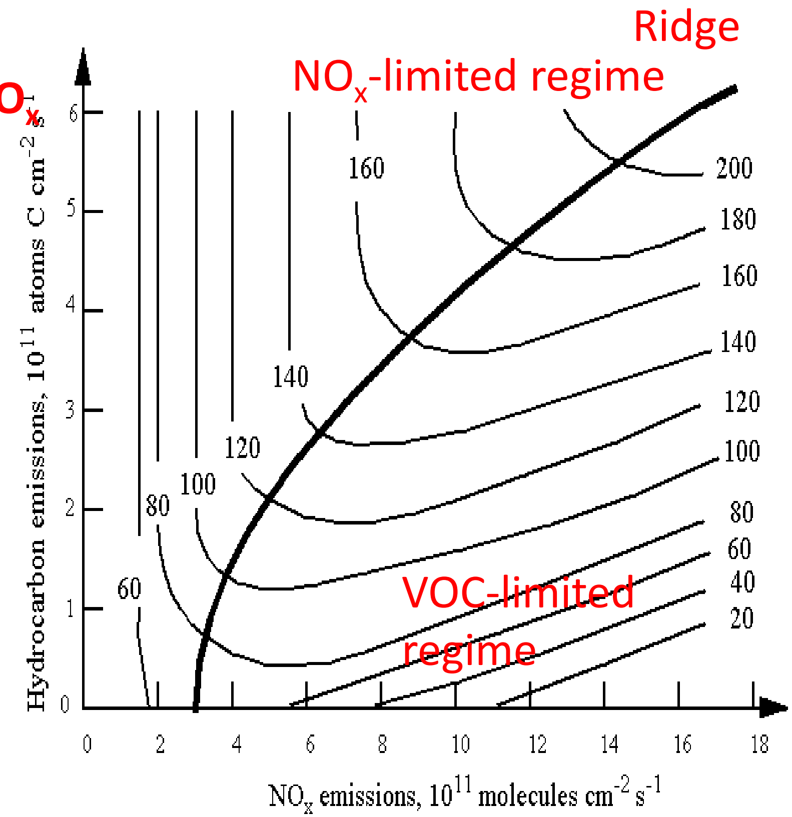
Chain termination

$$HO_2 + HO_2 \longrightarrow H_2O_2 + O_2$$

$$OH + NO_2 + M \longrightarrow HNO_3 + M$$

VOC limited (NO_x reactivity dominates)

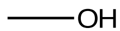
$$P_{O_3} = \frac{2k_{(VOC+OH)}P_{HO_x}n_{VOC}}{k_{(OH+NO_2+M)}n_{NO_2}n_M}$$

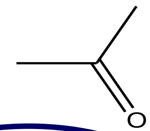


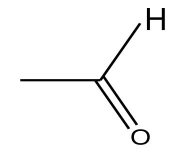
NO_x limited (VOC reactivity dominates)

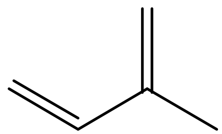
$$P_{O_3} = 2k_{(HO_2+NO)}\left(\frac{P_{HO_x}}{k_{(HO_2+HO_2)}}\right)^{1/2} n_{NO}$$

Some VOCs Present in Ambient air and Examples of Direct Health Impacts

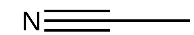
Methanol : 

Acetone : 

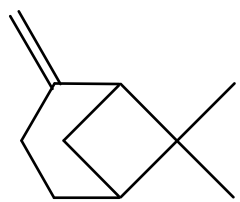
Acetaldehyde : 

Isoprene : 

ppbV-pptV levels

Acetonitrile : 

Butane : 

Pinene : 

Alkanes like ethane, propane

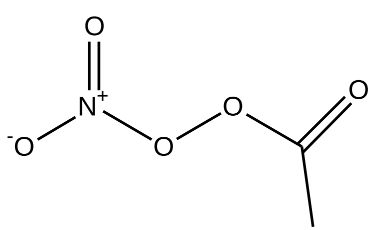
Alkenes like ethene, propene

Aldehydes like formaldehyde, glyoxal

Organic acids like acetic acid, formic acid

Aromatics like benzene, toluene, xylene

CARCINOGENIC (> 5ppbV)

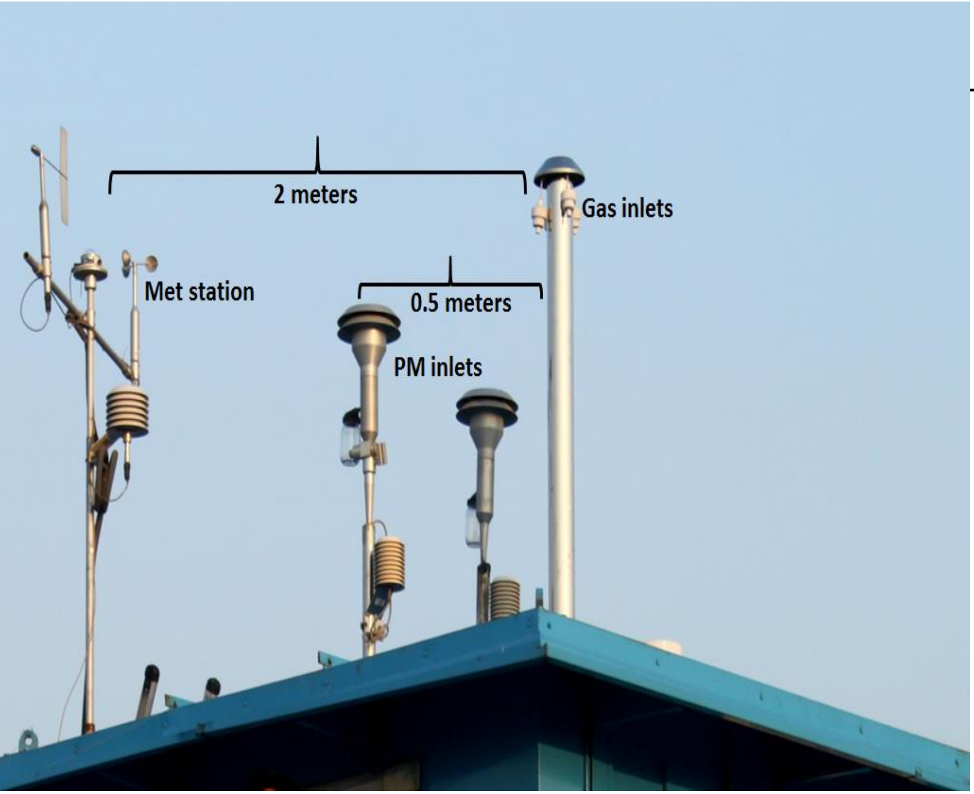
PAN : 

LACHRYMATOR

Direct Impacts on Health 

10,000 to 100,000 different compounds exist in the atmosphere

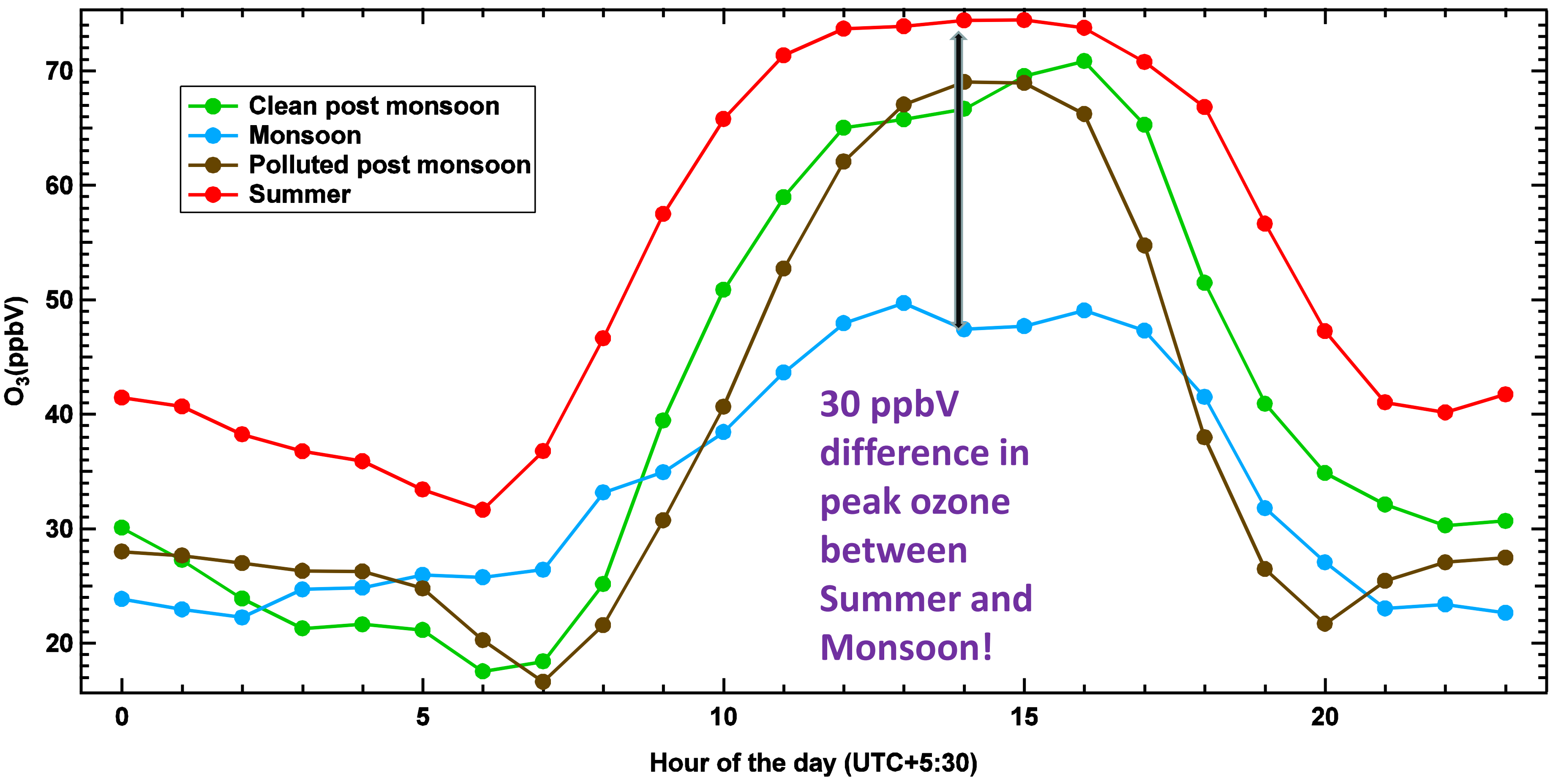
IISER Mohali Atmospheric Chemistry Facility



Measured species	TechnOique	Time resolution	Detection Limit
CO	Non-Dispersive Infra Red (NDIR)	1 min	30 nmol mol^{-1}
NOx	Chemiluminescense	1 min	150 pmol mol^{-1}
O_3	UV-Photometry	1 min	1 nmol mol^{-1}
SO_2	Pulsed UV-Fluorescence	1 min	1 nmol mol^{-1}
PM_{10}	β attenuation	1 min	< 4 $\mu g/m^3$
$PM_{2.5}$	β attenuation	1 min	< 4 $\mu g/m^3$
VOCs /OH Reactivity	PTR-MS	1 min	10-30 pmol mol^{-1} / 3 s^{-1}

V. Sinha*, V. Kumar and C. Sarkar , Atmos Chem Phys , 14, 5921-5941, 2014

Seasonality of Average Diel Ozone at Mohali



Bimodal Profile Driven by Dynamics and Emission Activity

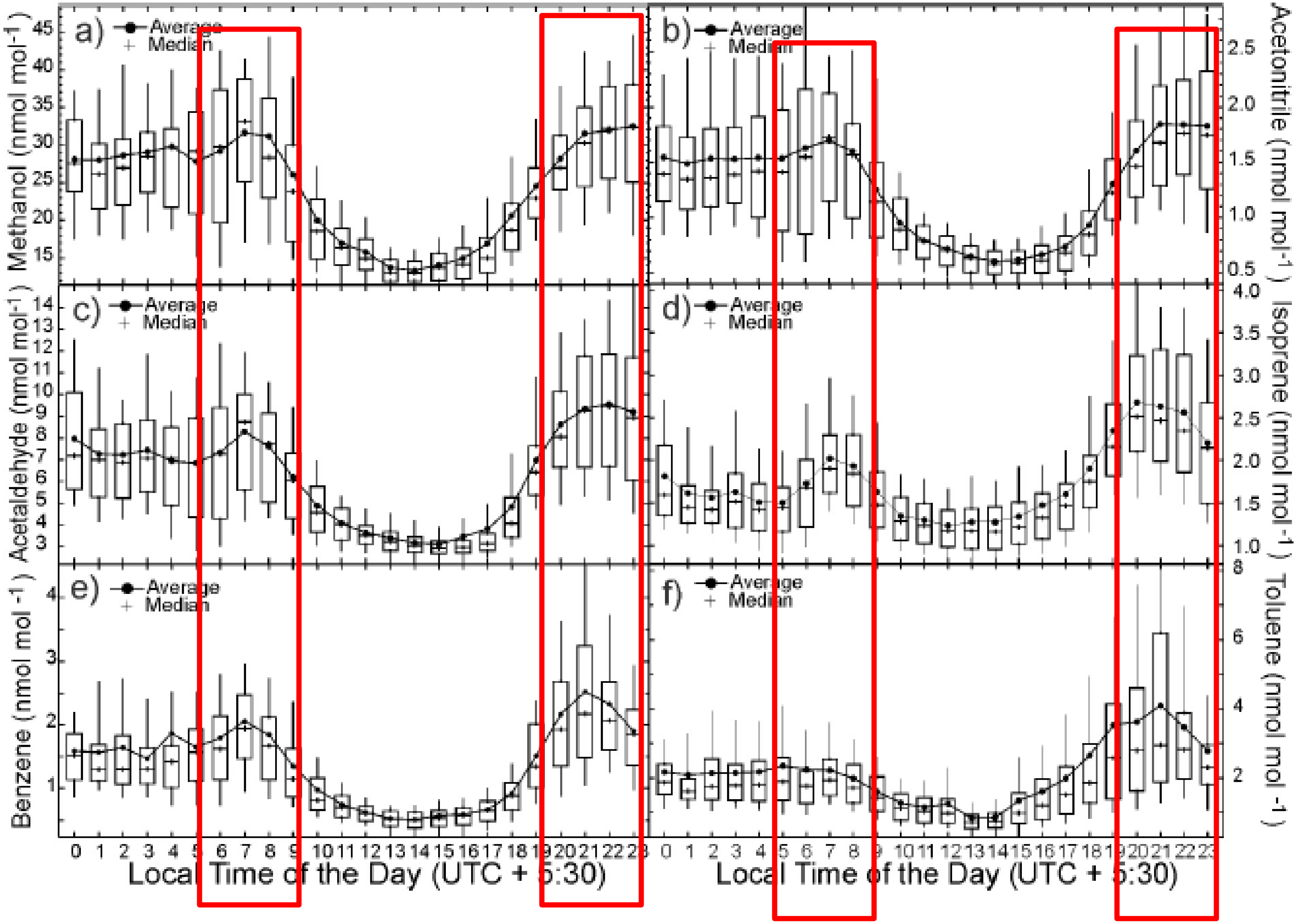
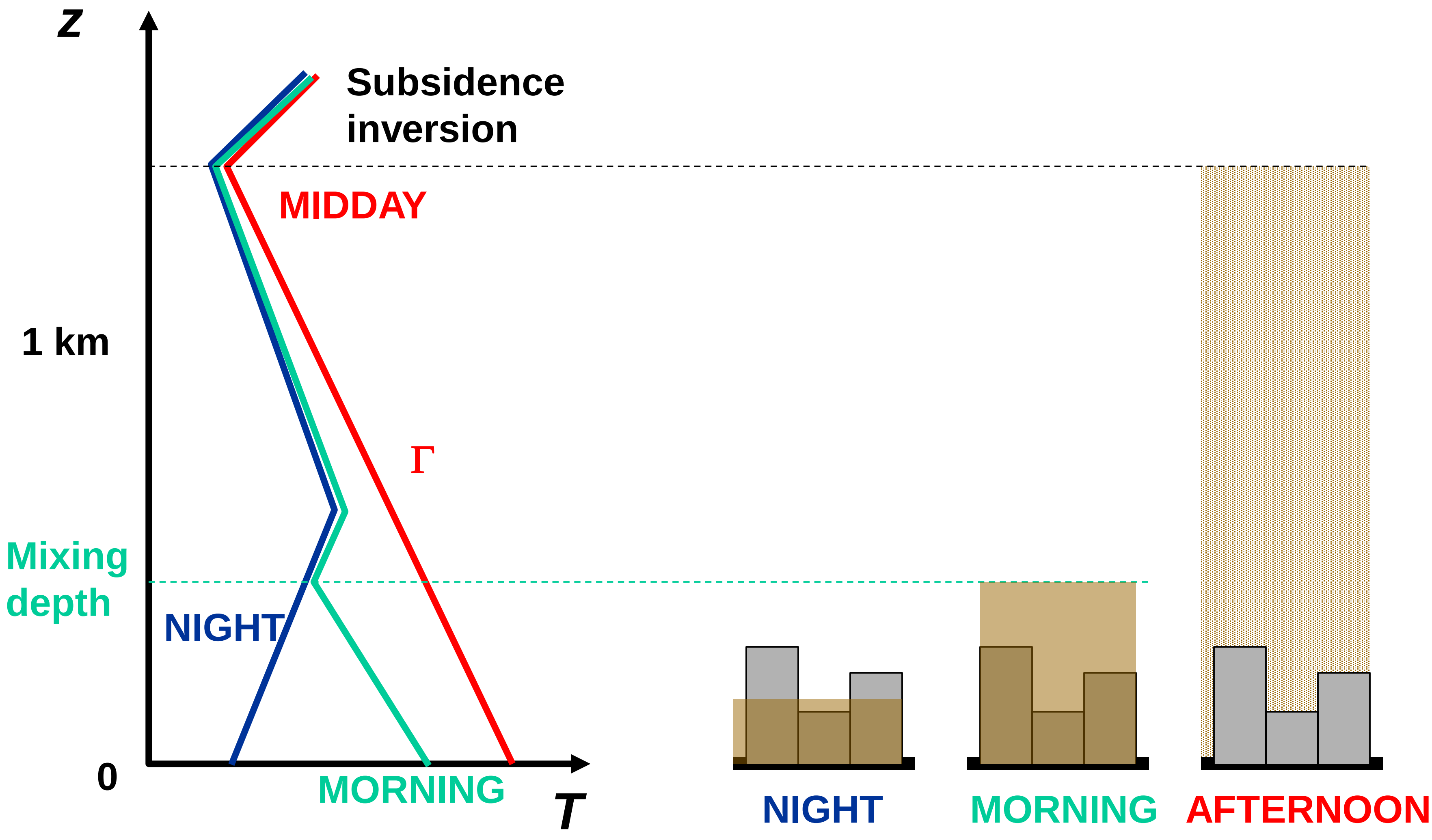


Fig. 9. Diel box and whisker plot of **(a)** methanol, **(b)** acetonitrile, **(c)** acetaldehyde, **(d)** isoprene, **(e)** benzene and **(f)** toluene derived from all measurements ($n > 14\,300$ for each species) in May 2012 at the measurement site.

Sinha et al., Atmos. Chem. Phys. Discuss, 2013

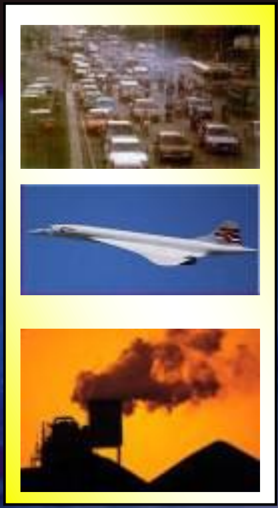
DIURNAL CYCLE OF SURFACE HEATING/COOLING: ventilation of urban pollution



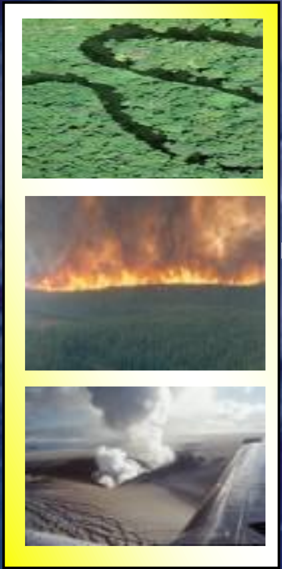
Overview

Sources

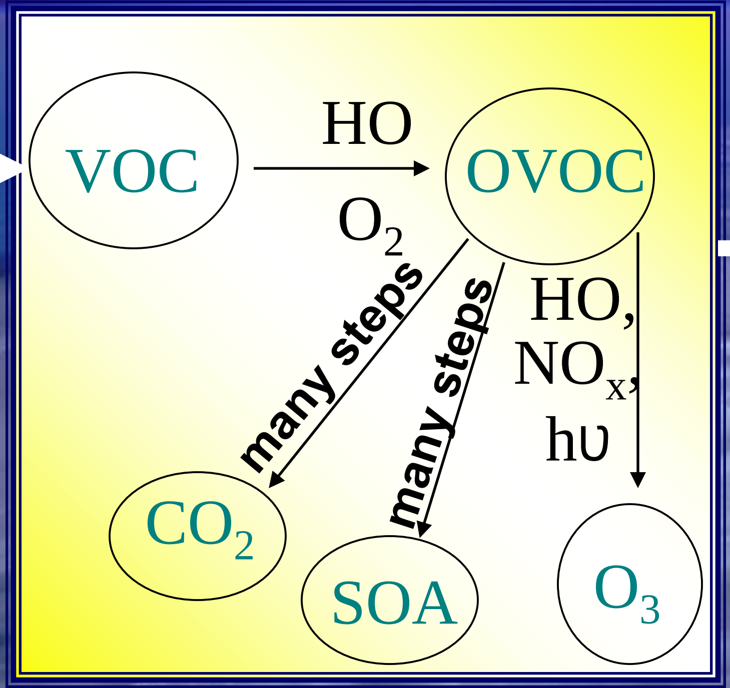
A n t h r o p o g e n i c



B i o g e n i c



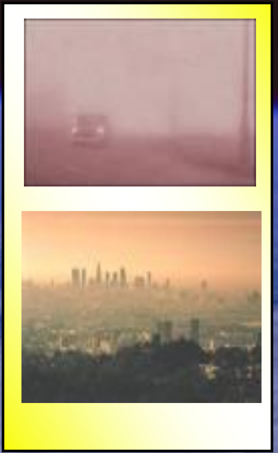
Dominant Chemical Processing



Primary Emissions (Gases and Aerosol)

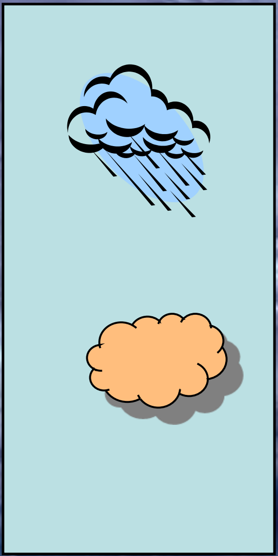
Approx. 1300 TgC / yr from NMHCs alone (Goldstein and Galbally, Env. Sci. & Tech, 2007)

Impacts



AIR QUALITY

HEALTH



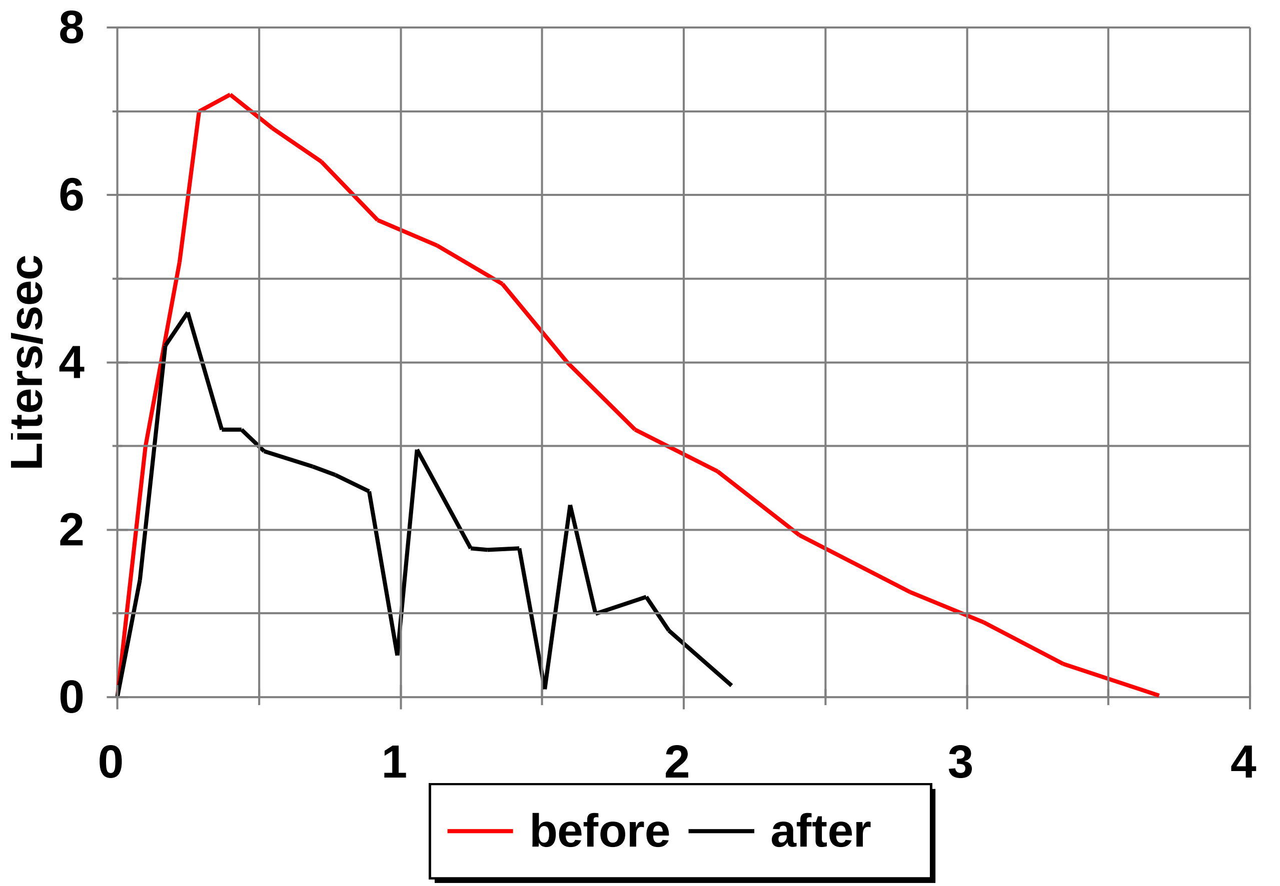
CCN & AEROSOL

CLIMATE

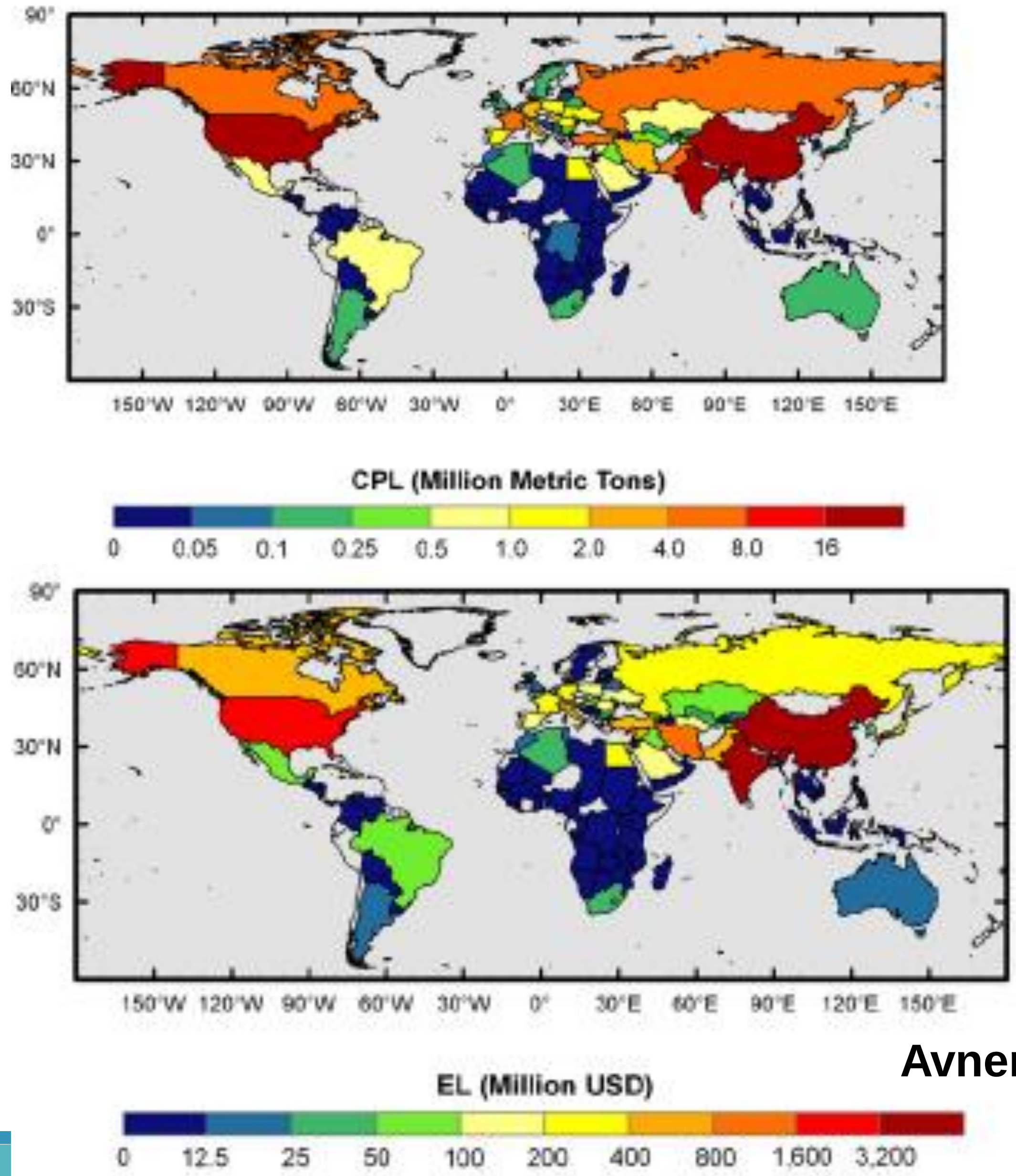
OZONE HEALTH EFFECTS

- Ozone causes dryness in the throat, irritates the eyes, and can predispose the lungs to bacterial infection.
- It has been shown to reduce the volume or the capacity of air that enters the lungs
- School athletes perform worse under high ambient O_3 concentrations, and asthmatics have difficulty breathing
- The current US standard has been just reduced from 0.12 ppm for one hour to 0.08 ppm for one hour

Lung function after exposure to O.32 ppm O_3



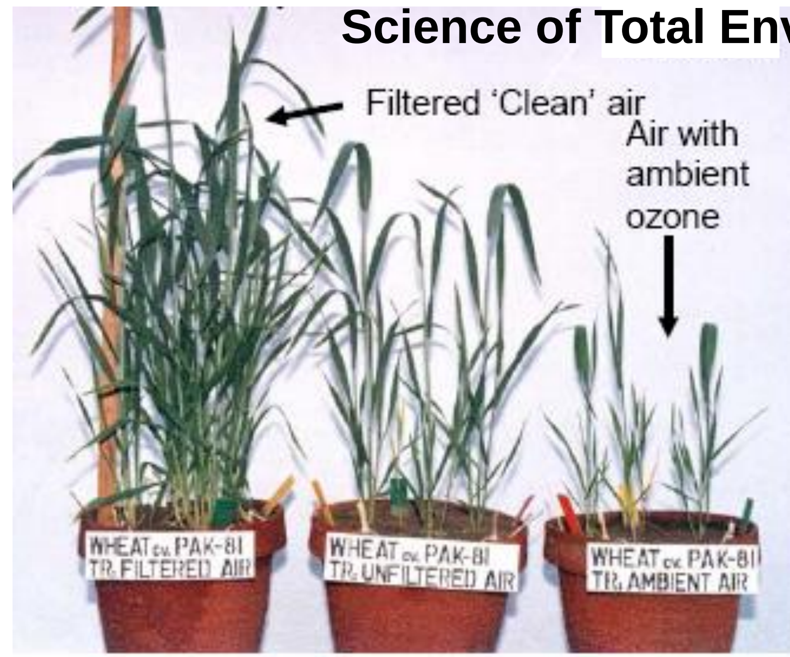
Crop, Economic Losses in 2000 due to AOT40



Crops: Wheat, maize, soyabean

Does not even include paddy !

A. Wahid , 2006
Science of Total Env.



Avnery et al., 2011, Atm. Env.